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Notice on Issuing Zhengzhou's "14th Five-Year" Ecological and Environmental Protection Plan

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Zhengzhou Municipal People's Government Office

About the issuance of Zhengzhou's "Fourteenth Five-Year" Ecological and Environmental Protection Plan

Planning Notice

Zhengzhengbanÿ2022ÿNo.42

To the management committees of all development zones, the people's governments of all districts, counties (cities), all departments of the municipal people's government, and all relevant units:

The "Zhengzhou 14th Five-Year Plan for Ecological and Environmental Protection" has been approved by the municipal government and is now issued to you. Please implement it conscientiously.

Zhengzhou

Zhengzhou City's "Fourteenth Five-Year" Ecological and Environmental Protection Plan

In order to promote the high-quality development of the city's economy and the high-level protection of the ecological environment, in accordance with the "Notice of the State Council on Issuing the 14th Five-Year Plan for Ecological and Environmental Protection" (Guofa [2021] No. 31), the Henan Provincial Notice on Issuing the 14th Five-Year Plan for Ecological Environmental Protection and Ecological Economic Development in Henan Province (Henan Government [2021] No. 44) and Notice of the People's Government of Zhengzhou on Issuing the 10th Five-Year Plan for National Economic and Social Development in Zhengzhou This plan was formulated in accordance with the "Notice on the Outline of the Long-Term Goals for 2035" (Zhengzheng [2021] No. 12).

1. Basic Situation

1. Current situation

Since the 13th Five-Year Plan, Zhengzhou City has earnestly studied and implemented Xi Jinping's thoughts on ecological civilization, guided by the concept of green development, focusing on serving high-quality development, and improving the quality of the ecological environment. We have made great efforts to fight the three major battles to protect blue skies, clear waters and clean land. The phased goals and tasks of the pollution prevention and control battle have been successfully completed. The quality of the ecological environment has been significantly improved. The people's sense of gain, happiness and security has been greatly improved.

The quality of the ecological environment has improved significantly. The quality of the atmospheric environment has improved significantly. In 2020, the proportion of days with good air quality in Zhengzhou reached 63%, an increase of 92 days compared with 2015, and the increase ranked third in the province. The annual average concentrations of PM10 and PM2.5 decreased by 49.7% and 46.9% respectively, and the improvement rates ranked first in the province. The city also ranked first in the ranking of air quality among the 168 key cities in the country.

We have successfully completed the goals of the 13th Five-Year Plan and the three-year action plan to win the battle for blue skies, and achieved the best results since the battle against air pollution. The quality of surface water environment continues to improve significantly, and national and provincial control The water quality of the city has reached Class III and above, the proportion of Class I to Class III water bodies has reached 75%, the black and odorous water bodies in the urban built-up areas have been completely eliminated, and all the centralized drinking water sources at the municipal level have reached Class III and above water quality. The safe utilization rate of polluted cultivated land and construction land has reached 100%, the soil pollution prevention and control system has been gradually improved, and soil environmental risks have been basically controlled.

The achievements in pollution prevention and control are outstanding. The city has achieved "zero" non-electric coal-fired boilers and dynamic "zero" scattered coal in plain areas, and has shut down 2.27 million kilowatts of coal-fired power units. Coverage, gas boilers with a steam capacity of more than 1 ton have basically completed low-nitrogen transformation. Deep treatment of the cement industry, elimination of high-polluting vehicles, monitoring of heavy-duty diesel vehicles, pollution control of non-road mobile machinery, and lifting of construction sites Leading the way in the country. Completed the regional demarcation and remediation of 53 "thousand tons per million people" drinking water sources, completed the investigation and remediation of risk sources in the protection area of the middle route main canal of the South-to-North Water Diversion Project; took the lead in the province in carrying out sewage treatment The district has basically achieved full collection and treatment of domestic sewage, and the centralized treatment rate of urban sewage in the district, county (city) has reached more than 95%. Completed a detailed investigation of the soil pollution status of agricultural land in the city, and completed the soil pollution status of land used by key industries and enterprises. Leading the province; taking the lead in the province to carry out basic groundwater environmental investigation and assessment, and preliminarily understand the groundwater pollution status in the central urban area.

Help high-quality development to make positive progress. Define "three lines and one list" and implement ecological environment zoning control. Carry out "per mu hero" comprehensive evaluation work, accelerate industrial structure adjustment, and reduce steel production capacity by 100 million tons. 1.01 million tons of mud, 510,000 tons of electrolytic aluminum, 20.38 million tons of coal, 503,500 tons of paper, 1.953 million tons of carbon, and 350,000 tons of concrete; 10,769 "scattered, disorderly and polluting" enterprises were rectified and banned, achieving dynamic "zero clearance".

We formulated the "Zhengzhou Municipal Bureau of Ecology and Environment's List of Minor Ecological and Environmental Violations Exempt from Penalties (Trial)", which was commended by the Ministry of Ecology and Environment. We launched the "Ecological and Environmental Enterprise Service" activity and deepened

To address the issue, all 53 administrative approval and service items of the Municipal Ecological Environment Bureau have been realized online, and the city has been promoted to the top 100 cities with the most friendly business environment in the world, and has successfully established itself as a model city for the construction of a national social credit system.

Ecological green creation presents a new look. Carry out the construction of the core demonstration zone for ecological protection and high-quality development in the Yellow River Basin, and implement the comprehensive improvement project of the Yellow River beach area. During the "Thirteenth Five-Year Plan" period, a total of 10,000 ecological corridors were built.

The national land greening area is 655,100 mu, the new green area is 25,852,600 square meters, and the sponge city construction area is 110.8 square kilometers. By the end of 2020, the forest coverage rate will increase to 35%, and the green area rate will be

The areas reached 36.1% and 13.79 square meters respectively. One national ecological civilization demonstration county, two provincial ecological counties, four provincial ecological towns, 18 provincial ecological villages, and municipal ecological

The 11th China (Zhengzhou) Garden Expo was held and the city was awarded the titles of "National Model Greening City" and "National Ecological Garden City".

Environmental governance capabilities continue to improve. The "Implementation Opinions on Accelerating the Construction of a Modern Environmental Governance System" was issued, and the construction of a modern environmental governance system was fully launched.

Zhengzhou City's second national pollution source census was recognized by the state, achieving full coverage of fixed pollution source discharge permits.

We will improve the quality of water environment and air quality and carry out ecological compensation. We will actively carry out rural environmental quality monitoring and basically achieve full coverage of the automatic environmental quality monitoring network. We will continue to promote the optimization of the ecological environment big data platform.

We will promote the upgrading and improvement of the hazardous waste Internet of Things system, explore the pilot work of hazardous waste collection from small-scale waste-producing units, and build a hazardous waste Internet of Things supervision system that is at the forefront of the country.

(II) The current situation

During the 14th Five-Year Plan period, the ecological environment continued to improve and faced many opportunities and favorable conditions. Xi Jinping's ecological civilization thought has taken root in the hearts of the people, the new development concept has been deeply implemented, and the dividends of ecological civilization system reform have continued.

The joint efforts to protect the ecological environment have been further enhanced, providing a solid foundation for the continuous improvement of the ecological environment; the country has actively responded to climate change, focusing on carbon reduction as a strategic direction, promoting the synergy of pollution reduction and carbon reduction, and promoting economic

Green transformation provides an action guide for ecological and environmental protection; the implementation of the two major strategies of ecological protection and high-quality development of the Yellow River Basin and promoting high-quality development in the central region provides a basis for coordinating economic and ecological development.

An important opportunity; Zhengzhou is comprehensively building a modern national central city and accelerating its progress towards an international metropolis, providing new impetus for the city's new work in ecological and environmental protection.

During the 14th Five-Year Plan period, environmental pollution prevention and control work entered a new stage, and the improvement of ecological environment quality entered a critical period from quantitative change to qualitative change. Compared with the construction of a beautiful China, Zhengzhou still has the following shortcomings:

First, there is still a gap between the ecological environment quality and the target. The concentrations of PM2.5, PM10 and ozone have not yet reached the second-level standard of atmospheric environment quality, and the ozone concentration is relatively high, which has a greater impact on air quality.

There is still the phenomenon of direct discharge of sewage, and the municipal-controlled sections have not yet fully met the assessment requirements. Black and smelly water bodies occasionally appear in rural areas.

Second, the structural pollution contradiction is more prominent. The city still has many traditional industries such as cement, refractory materials, bricks and tiles, and reinforced concrete. There are still a large number of high-energy-consuming, high-emission, and low-output enterprises.

The city's motor vehicle ownership has exceeded 4.3 million, ranking sixth in the country, and is still growing rapidly at a rate of 400,000 vehicles per year; the heavy trucks brought by the city's status as a transportation hub and the concentration of wholesale markets

Third, there are shortcomings in environmental infrastructure construction and urban management. Rainwater and sewage mixing still exists in some areas of the central city, and the aging of drainage pipes is common. Some of the rural sewage treatment facilities that have been built have been

The pollution control benefits have not been fully realized. The waste incineration, sludge disposal and hazardous waste disposal capacities are close to saturation, and the domestic waste sorting center and recycling system have not been fully established.

The control effect still needs to be further improved.

Fourth, a modern environmental governance system has not yet been established. A multi-party co-governance environmental governance system led by the government, with the participation of enterprises, social organizations and the public needs to be accelerated and improved.

Monitoring capabilities still need to be strengthened, the supervision and law enforcement tasks are heavy, and the construction of environmental protection smart platforms is still lagging behind.

II. General Requirements

1. Guiding ideology

Guided by Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era, we will fully implement the spirit of the 19th CPC National Congress and the 19th CPC Central Committee, and thoroughly implement Xi Jinping Thought on Ecological Civilization and General Secretary Xi Jinping's

We must follow the instructions of the speech, stand firm in the new development stage, implement the new development concept, actively integrate into the new development pattern, accelerate the construction of national central cities, and deepen the ecological protection and high-quality development strategy of the Yellow River Basin.

The fundamental goal is to maintain the growing beautiful ecological environment, adhere to the general tone of work of seeking progress while maintaining stability, grasp the general requirements of reducing pollution and carbon emissions, fight hard to prevent and control pollution, and accelerate the promotion of the ecological environment governance system.

We will continue to enhance the ecological advantages of central cities, build a green foundation for the beautiful Central Plains, and lay a solid ecological foundation for building a beautiful Zhengzhou where people and nature live in harmony.

(II) Basic Principles

People first, ecology benefits the people. Relying on the people and serving the people, we will focus on solving prominent ecological and environmental problems around the people, create a good production and living environment for the people, and provide more high-quality

To meet the people's growing need for a beautiful ecological environment.

Ecological priority and green development. Establish the concept that green mountains are gold and silver mountains, implement the sustainable development strategy in depth, give full play to the leading and forcing role of ecological and environmental protection, and accelerate the formation of green ecology.

Promote the comprehensive green transformation of economic and social development.

Demonstrate innovation and highlight characteristics. Focusing on ecological creation and waste-free city construction, we will enhance the demonstration and innovation capabilities of central cities, reflect the characteristics of system and innovation-driven development, and explore ecological and environmental characteristics with local characteristics.

Improve the overall coordination mechanism in the field of ecological civilization, stimulate the internal motivation of various responsible entities such as party committees, governments, enterprises, and the public, improve the ecological environment supervision system, and form

Build a modern environmental governance system with governance capabilities that are adapted to environmental management needs.

(III) Main objectives

By 2035, the ecological system will be fully optimized, the ecological environment quality will be excellent, the ecological service functions will be complete, the quality of urban and rural life will be excellent, the ecological economy will be green and efficient, the ecological culture will be prosperous and harmonious, and the sky will be blue and white.

It basically meets the people's needs for a beautiful ecological environment and builds a beautiful Zhengzhou where people and nature live in harmony.

By 2025, the ecological environment will continue to improve, structural adjustment will be advanced in depth, green and low-carbon development and green living standards will be significantly improved, the ecosystem will be more stable, and the ecological awareness of the whole society will be significantly increased.

The modernization of the system and governance capabilities has been initially achieved, ecological and environmental protection has reached a new level, and the construction of a beautiful Zhengzhou has made significant progress.

Ecological protection. Strengthen ecological protection and restoration of the Yellow River Basin, build a core demonstration zone for ecological protection and high-quality development in the Yellow River Basin, and ensure that the ecological protection red line area of the entire city does not decrease and the ecological quality index does not increase.

Green development. The green transformation of production and lifestyle has achieved remarkable results. The total emission of major pollutants has continued to decrease. The proportion of high-energy, high-water-consuming and high-polluting industries has been further reduced. Energy consumption per unit of regional GDP has been reduced.

Reduce the ratio to achieve provincial targets.

Environmental improvement. The progress of meeting air quality standards has accelerated, with the annual average concentration of PM2.5 dropping to 40 micrograms per cubic meter, and heavy pollution has been basically eliminated; black and odorous water bodies in county-level urban built-up areas have been basically eliminated, and river and lake water quality has improved.

The soil and groundwater environmental quality remained stable, with some improvements in some areas.

Improved governance. The reform of the ecological civilization system has been deeply implemented, the ecological environment protection infrastructure has basically achieved full coverage in urban and rural areas, the shortcomings of ecological environment governance capacity have been quickly made up, and the ecological civilization awareness of the whole society has been significantly improved.

The effectiveness of environmental governance has been further improved.

Column 1 Main indicators of ecological and environmental protection in Zhengzhou during the 14th Five-Year Plan period

Category	No.	index	2020 (baseline value)	2025 target	indicator nature	
ring territory quality quantity change good	1	Annual average concentration of PM2.5 (ÿg/m3) ÿÿÿ	51	40	Binding	
	2	Ratio of days with good air quality ÿÿÿ	63.0	66.0	Binding	
	3	Percentage of water quality that meets national and provincial control standards ÿÿÿ	87.5	100	Binding	
	4	Ratio of Class V water in national groundwater control points example(%)	–	0	Anticipation	
	5	Centralized drinking water at county level and above The water quality at the source reaches or exceeds Class III Proportion (%)	90.9	100	Binding	
	6	Black and smelly water bodies in county-level urban built-up areas Proportion (%)	–	Basically eliminate the	expectation	
	7	Rural domestic sewage treatment rate ÿÿÿ	Approx. 50	70	Anticipation	
green ÿ Low carbon hair exhibition	8	Unit GDP Carbon emission reduction ratio (%)	–	Completion of provincial	target binding	
	9	Energy consumption per unit of GDP Low rate (%)	–	Completion of provincial	target binding	
	10	Total water consumption in the city (billion cubic meters) ÿÿ	20.7365	26.02	Binding	
	11	Water consumption per 10,000 yuan of GDP Decrease rate (%)	–	Completion of provincial	target binding	
Sewage dye thing Row put total quantity control system	12	Nitrogen oxide emission reductions from key projects (Ton)	–	[19075]	Constraint	
	13	Key projects for reducing volatile organic compounds Displacement (tons)	–	[10327]	Constraint	
	14	Key reductions in chemical oxygen demand (Ton)	–	[10434]	Constraint	
	15	Ammonia nitrogen emission reduction in key projects (Ton)	–	[263]	Constraint	
ring territory ÿ risk	16	Safe utilization rate of polluted cultivated land ÿÿÿ	–	95	Binding	
	17	Safe Utilization of Key Construction Land	–	Effectively ensure	binding	
	18	Hazardous waste utilization and disposal rate	100	100	Anticipation	

Defend control		ÿÿÿ			
	19	Medical services in built-up areas of cities above county level Waste harmless disposal rate (%)	100	100 Anticipation	
green ÿ ÿ state	20	Ecological Quality Index (EQI)	–	Stable and positive expectations	
	21	Forest coverage rate (%)	35	35 Binding	
	22	Ecological protection red line area (10,000 square meters Square kilometers)	–	No reduction in constraints	
Note: [] refers to the cumulative value over five years					

3. Strengthen ecological protection of the Yellow River and improve the quality of the ecosystem

1. Build a core demonstration zone for ecological protection and high-quality development in the Yellow River Basin

Optimize the ecological protection space along the Yellow River. Along the south bank of the Yellow River, east to Zhongmu Langchenggang Town, west to Gongyi Kangdian Town, south to Lianhuo Expressway, north to the middle line of the main channel of the Yellow River, with an area of about 1,200 square kilometers. The core demonstration area of protection and high-quality development. East to Zhengzhou Huayuankou East 107 National Road, west to Zhengyun Expressway connecting Tianlun Road (including Guangwu Mountain), south to Lianhuo Expressway connecting Suoxu River (with the southern bank line The Fourth Ring Road, north to the middle line of the main Huanghe River, covers an area of about 210 square kilometers to build the starting area. Coordinate and optimize the ecological, agricultural and urban space, strictly adhere to the three control lines, and scientifically determine the protection and development pattern of the core demonstration area. A new type of urban and rural system with complementary and integrated development and a town space system with a moderate layout density and co-prosperity between rivers and cities.

Continue to clean up and manage river beaches. Promote flood control in the river basin, river channel and beach area management, and innovate the "three-beach management" of shallow beach, middle beach and high beach. Continue to clean up the chaos on the river beaches, and completely prohibit the main stream of the Yellow River and the first We will crack down on all types of illegal construction within the scope, accelerate the construction of a digital ecological protection monitoring platform, and carry out special rectification work on ecological environment protection along the Yellow River on a regular basis.

Improve the construction of ecological corridors along the Yellow River. Improve the construction level of ecological corridors along the south bank of the Yellow River in Zhengzhou, and build windbreak, sand fixation, and embankment protection forest belts. Rationally select and match native tree species, precious tree species, and special The tree species are arranged in a three-dimensional manner, and the tree species, coniferous and broad-leaved tree species are configured to make the forest phase transition from evergreen to bright and colorful, thereby improving the diversity of the landscape.

Restore the ecological function of the Yellow River Wetland. Relying on the wetland resources such as the Yellow River main stream, reservoirs, lakes, and swamps, carry out ecological restoration and repair of wetlands with weakened functions and ecological degradation. The project aims to accelerate the construction of wetland parks along the Yellow River, with the protection and restoration of natural wetlands as the core. The wetland quality will be further improved through ecological restoration measures, and the ecological function of the Yellow River wetland will be systematically restored.

(II) Create a model of water-saving city

Enhance water resource support and guarantee capabilities. Adhere to the priority of water conservation and return water to the river, implement the principle of determining cities, land, people and production based on water, and implement the strictest water resource management and water extraction permit system. Optimize the water resource allocation pattern and improve the allocation efficiency by combining groundwater, natural water and recycled water, local water and external water, conventional water and unconventional water. Actively strive for the South-to-North Water Diversion Water Rights Trading Index and the South-to-North Water Diversion Water Rights Trading Index. The goal is to make resourceful use of the water returned from the South-to-North Water Diversion Project and to improve the level of water conservation and intensive use of the Yellow River.

Implement deep water conservation and water control actions. Accelerate the implementation of high-efficiency water-saving irrigation projects and agricultural water conservancy project construction, actively promote pipeline water irrigation, sprinkler irrigation, micro-irrigation and other high-efficiency water-saving irrigation technologies, organize and carry out Pilot projects. Promote water-saving livestock and fishery breeding technologies to improve water efficiency in the breeding industry. Strengthen the management of the entire industrial water use process, improve the water supply and use metering system and online monitoring system, and vigorously promote advanced water-saving technology. Water balance tests, water use audits and water efficiency benchmarking will be carried out regularly for key enterprises, and water-saving transformation will be implemented in a classified and phased manner within a time limit for enterprises that exceed the water quota standards. By 2025, the water efficiency coefficient of farmland irrigation will be improved. The water consumption per industrial value added shall not exceed 10.3 cubic meters.

Strengthen urban and rural water conservation and loss reduction. Implement regional metering management of urban water supply pipes and "one meter per household", and renovate old water supply pipes. Promote the popularization of water-saving appliances in daily life, and carry out water conservation in government agencies, schools, hospitals, etc. Strictly control water consumption in high-water-consuming service industries, and continue to create water-saving enterprises, units, and communities. Improve rural centralized water supply and water-saving supporting facilities, promote the use of water-saving appliances, and gradually implement metered charging. Runners Action will further implement water efficiency labeling, water conservation certification and contract water conservation management. By 2025, the leakage rate of urban pipe networks will be reduced to 9%.

Promote unconventional water utilization. Continue to promote the utilization of unconventional water resources such as rainwater, mine water and sewage. Newly built, renovated and expanded sewage treatment plants with a design capacity of more than 50,000 cubic meters per day should be equipped with Expand the use of recycled water, incorporate recycled water into the water resource allocation system, and allocate recycled water to units with an annual water consumption of more than 5,000 cubic meters within the coverage of the recycled water pipeline network. By 2025 The recycled water utilization rate reached 30% and the county (city) built-up area reached 25%.

Control of groundwater over-exploitation areas will be carried out. Groundwater pressure extraction will be strictly implemented. By 2022, the city's groundwater will basically achieve a balance between extraction and replenishment. During the "Fourteenth Five-Year Plan" period, the total amount of groundwater pressure extraction in the city will reach 75 million cubic meters. Improve the dual control system of groundwater volume and water level, build a network of automatic groundwater level monitoring stations in Zhengzhou, and construct a groundwater level monitoring and early warning forecasting system to realize dynamic monitoring of groundwater.

3. Strengthening the protection of ecosystems and biodiversity

Improve the construction of ecological security pattern. With the Yellow River Basin and Songshan Mountain Area as the framework, build a large ecological security pattern of "one river, one mountain, two wings, five waters, six wedges, and seven corridors". Restore the Yellow River beach area and lateral seepage area The ecological system will be integrated and the ecological landscape belt dominated by "tidal flat wetland-landscape protection forest-ecological agriculture" will be gradually improved. In the Songshan-Fuxishan rocky mountain area, quarrying and mining projects will be gradually shut down and ecological restoration will be carried out. Utilize, vigorously develop coniferous and broad-leaved mixed forests, and strengthen the ecological conservation of forest vegetation. Rationally promote water expansion and humidification projects, build regional oxygen sources and ventilation corridors to improve urban microclimate. Urban and rural construction should combine ecology, It is advisable to control the development intensity and spatial form, and reserve blank areas for ecology and development in urban space.

Create an ecological network that integrates urban and green areas. Connect urban public spaces, parks, and scenic spots through ecological corridors, build urban waterfront landscape recreation belts and urban greenway systems, and form a "small cycle + large The urban green space system creates an integrated ecological network that combines urban and green areas. Create shared urban waterfronts on the Jialu River and Suoxu River, and build a rich three-dimensional waterfront landscape based on the Jinshui River, Xiong'er River, and Dongfeng Canal. The "four core, six belts and multiple green corridors" system will carry out in-depth construction of the systematic and connected ecological corridors through the construction of "greening, beautification, colorization and optimization", and comprehensively improve the quality of ecological corridors.

Strengthen ecosystem protection and restoration. Scientifically define the scope and zoning of nature reserves, and accelerate the integration and optimization of various types of nature reserves. Implement wetland protection and Restore the ecological function of wetlands. Expand wetland area through multiple channels such as relocation of beach area residents, quality improvement of 100 cities, and water system management. Carry out aquatic vegetation restoration, water level regulation, eutrophication control, and alien invasive species control.

Restore comprehensive management and gradually improve the quality of wetland ecosystems. Strengthen farmland shelterbelt networks, desertification management and degraded forest restoration to build a plain ecological green network. Vigorously promote the greening of Mangling and afforestation in difficult areas.

Improve the level of restoration and comprehensive management of the mining geological environment, and carry out restoration and management of historical mines in accordance with the principle of "planting forests where suitable for forests, and planting land where suitable for cultivation".

The principle of "creating scenery where it is suitable" should be followed and all possible measures should be taken to prevent and control land loss. By 2025, the newly added area for land loss prevention and control will be 274 square kilometers, and the governance rate of the historical mining geological environment left over in the "three zones and two lines" will reach 90%.

Strengthen biodiversity protection. Coordinate in situ and ex situ conservation, strengthen the protection and restoration of nationally protected and rare and endangered wild animals and plants and their habitats and original habitats, and connect important species migration routes.

Focusing on biodiversity-rich areas such as Songshan Mountain and Huanghe Wetland, we will further promote the background survey, monitoring and assessment of biodiversity, establish and improve in situ protection.

Ex situ conservation facilities such as germplasm resource banks.

Strengthen biosafety management. Establish and improve a biosafety risk prevention and control and governance system, and strengthen system governance and full-chain prevention and control. Strengthen the control of invasive alien species and conduct a census of invasive alien species.

We will monitor and warn of invasive alien species in key areas such as farmland, fishery waters, forests, grasslands, and wetlands, and strengthen risk warning and emergency response at the coast. We will accelerate the establishment of a genetically modified organism environmental safety monitoring network and strengthen

Environmental safety supervision of technology.

(IV) Improving the level of ecological demonstration creation

Consolidate the achievements of the creation of a national ecological garden city. Accelerate the construction of a four-level park system consisting of suburban parks, comprehensive parks, community parks, and micro parks (small gardens), and promote the organic integration of park forms and urban space.

Cities. In the periphery of cities, relying on existing ecological and cultural resources such as mountains, woodlands, water systems, wetlands, natural villages, and historical and cultural heritage sites, we will continue to promote the construction of suburban parks. Xiliu Lake Park was built.

40 comprehensive parks and specialized parks; accelerate the construction of 460 community parks and micro parks (small parks) in the central urban area; complete the removal of fences and greening and infrastructure construction of 15 parks such as Century Park and Moon Park

By 2025, the per capita park green space area in built-up areas will reach 14 square meters, the greening rate of built-up areas will reach 37.5%, and the coverage rate of park green space service radius will reach more than 93%.

Promote the establishment of demonstration areas for ecological and civilization construction. Actively promote the establishment of "Green Water and Yellow Mountains are Gold and Silver Mountains" practice and innovation bases and demonstration areas for ecological and civilization construction, and increase the establishment of demonstration areas for ecological and civilization construction in the Yellow River Basin.

By the end of 2022, Xinmi City will complete the construction plan of the "Green Water and Green Mountains are Gold and Silver Mountains" practice and innovation base, Gongyi City will complete the construction plan of the national ecological civilization demonstration county, and Xinzheng City will complete the provincial

Other development zones, districts, counties and cities shall revise or complete provincial ecological county construction plans.

Carry out forest city construction. Focus on the construction of urban forest networks and the improvement of quality and efficiency as well as the construction of forest ecological and cultural systems, and use idle land, barren hills and slopes around cities to carry out afforestation and build large areas of forests.

Zhongmu County, Xingyang City, and Xinzheng City will carry out the creation of provincial forest cities, and Dengfeng City, Gongyi City, and Xinmi City will carry out the creation of national forest cities.

Four forest cities were built, six towns with forest characteristics at or above the municipal level, and 200 forest villages were completed.

Create a practice base for the transformation of ecological product value. Improve the natural resource asset property rights system, promote the registration of natural resource rights, and conduct an ecological product information survey. Encourage all districts, counties, and cities to strictly protect

Under this circumstance, we will promote the realization of the value of ecological products in a scientific and rational manner by focusing on diversified models and paths such as ecological tourism, high-efficiency characteristic agriculture, leisure and health care, and creating distinctive regional public brands of ecological products.

Column 2 Key Ecological Protection and Construction Projects
1. Recycled water utilization. Matougang sewage treatment plant external recycled water pipeline project, recycled water utilization four-ring pipeline Engineering, Zhengzhou South China City sewage treatment plant water reuse project. 2. Wetland restoration and protection. Revise the Zhengzhou Yellow River Wetland Provincial Nature Reserve and Zhengzhou Yellow River National Wetland Park. Overall planning, Zhengzhou Huanghe Wetland Nature Reserve Smart Monitoring Platform Project, Zhengzhou Huanghe National Wetland Park Improvement Project, Zhengzhou Huanghe Wetland Zhongmu Bird Habitat Protection Area Construction Project, Jialu River and Lake Integrated Wetland Ecological Restoration Engineering, rare and endangered wildlife protection and rescue site projects. 3. Construction of country parks. Chaohu Country Park, Changzihu Country Park, Xingyang Suohe-Wanshan Country Park Park, Xinmi Yuanzhuang Country Park, Xinmi Qinshui River Country Park, Gongyi Liquan-Yiluo River Country Park, etc. 4. Construction of forest parks. Xinmi Xuehuashan Forest Sports Park, Dengfeng Xianshan Forest Park, Henan Songding National Jia Forest Park and Zhongmu Forest Park improvement projects. 5. Ecological corridor construction. Landscape improvement project of shelterbelts along the main canal of the middle route of the South-to-North Water Diversion Project, and the construction of the Jialu River —Protection forest landscape improvement project on both sides of the old route of the Grand Canal during the Sui and Tang Dynasties, and green ecological belt improvement project along the Beijing-Guangzhou Railway The green ecological belt improvement and upgrading project along the Beijing-Guangzhou high-speed railway and the Longhai Railway ecological belt upgrading project will continue to be promoted. Ecological corridors along 5 railway lines, 11 canals, 10 expressways, and 31 radial roads into the city Construction of unfinished parts of the project and upgrading and connecting of completed sections. Construction of the urban section of Provincial Highway 312 and the long 20 200-km ecological corridors such as Jiangjiang Road and Jiangang Road, and 20 300-km demonstration corridors such as the Third Ring Expressway and Jiangshan Road. Greenway improvement project.

4. Adhere to the guidance of innovative mechanisms and promote green and low-carbon development

1. Improve the green and clean development mechanism

Implement ecological and environmental zoning control. Establish resource and environmental carrying capacity, improve the "three lines and one list" ecological and environmental zoning control system, and establish a dynamic update and adjustment mechanism. Conduct ecological protection red line survey and demarcation.

The red line supervision system uses the ecological environment control unit and the ecological environment access list as the regional industrial layout, structural adjustment, resource development, urban construction, major project site selection, planning environmental assessment, and ecological

Implement the spatial control boundaries of ecological protection, basic farmland, and urban development, and promote intensive, green, and low-carbon urban development.

Continue to optimize industrial layout. Promote the withdrawal of key polluting enterprises that do not conform to urban construction plans, industry development plans, ecological and environmental functional positioning, or involve hazardous chemicals and other environmental risks from urban construction.

Polluting enterprises will be relocated out of the city. Each county (city) will further sort out the heavily polluting enterprises in the built-up areas and formulate and implement annual work plans. It is strictly prohibited to build new "two highs" within a certain range along the banks of the main stream of the Yellow River and its main tributaries.

Industrial parks, continue to promote the optimization of layout and structural adjustment of high water consumption, high pollution and high risk industries in the Yellow River Basin. Continue to improve the agglomeration level of industrial parks such as casting, refractory materials, and agricultural and sideline food processing.

Optimize and upgrade special action plans, and gradually complete the relief, relocation, transformation and quality improvement of urban commercial markets.

Deepen the reform of the "per mu hero" assessment. In accordance with the "Implementation Measures for the Classification and Comprehensive Evaluation of Industrial Enterprises in Zhengzhou", organize and implement the classification and comprehensive evaluation of industrial enterprises with districts, counties (cities) and development zones as the main body.

The results will be promptly announced to the public and used as an important basis for differentiated policies. Strictly implement comprehensive evaluation and optimization of factor allocation measures, and lay a good foundation for taxation and water use in accordance with the principle of "high utilization efficiency and high factor supply".

A combination of policies will be used to make good use of the reverse pressure mechanism, promote the rapid gathering of resources and factors to high-quality enterprises, and build a sound regional incentive and constraint mechanism.

(II) Promote regional industrial optimization and upgrading

Resolutely curb the blind development of "two high" projects. Strictly control the construction of high energy consumption and high emission projects. In principle, prohibit the construction and expansion of steel, electrolytic aluminum, cement, and aluminum for the sole purpose of increasing production capacity.

Strictly review project filings, strengthen on-site project inspections, and maintain a high-pressure posture to crack down on illegal new capacity projects. Construction projects should be carried out according to

Reduce requirements, implement equal or multiple substitution, and make the substitution plan and implementation status public.

Reduce energy consumption in industrial enterprises. Continue to carry out energy efficiency, water efficiency leaders and green manufacturing improvement actions, and focus on energy-saving technology transformation of high-energy-consuming equipment systems such as boilers, kilns, motors, fans, and internal combustion engines.

Clean, efficient water resource utilization and green basic manufacturing processes will be promoted to cultivate a group of energy efficiency and water efficiency benchmark enterprises. Clean production audits will be promoted in key industries to promote the combination of green technology and management.

The energy consumption of industrial added value decreased by about 4% annually.

Accelerate the elimination of existing excess production capacity. Continue to eliminate obsolete production process equipment and products; combine the "per-acre evaluation" and performance grading comprehensive evaluation to encourage and guide refractory materials, carbon, and cement.

Enterprises in polluting and high energy-consuming industries are required to exit.

Continue to investigate and rectify "scattered, disorderly and polluting" enterprises. Continue to maintain the dynamic elimination of "scattered, disorderly and polluting" enterprises, and implement measures such as closure, abolition, integration and relocation, rectification and improvement according to categories. Improve and implement the four-level provincial, municipal, county and township

Strengthen the main responsibilities of districts, counties (cities), towns (streets), strengthen environmental supervision and inspections, implement dragnet-style inspections and list-based, ledger-based, and grid-based management, and resolutely put an end to "scattered, messy, and polluting" enterprise projects.

"Scattered, disorderly and polluting" enterprises are resurrecting and moving to other places.

Accelerate the construction of a circular industrial system. Expand the comprehensive utilization of industrial bulk solid waste, promote the resource utilization of waste electronic and electrical products, power batteries, etc., and continuously improve resource utilization efficiency.

Improve the efficiency of industrial park utilization, promote comprehensive energy transformation, third-party pollution control and green transformation. Accelerate the construction of national-level industrial resource comprehensive utilization demonstration bases, support the establishment of renewable resource comprehensive utilization demonstration enterprises

Standardize the recycling and disposal system. By 2025, the comprehensive utilization rate of industrial solid waste will reach more than 80%.

Strengthen the energy-saving and environmental protection industry. Give full play to Zhengzhou's industrial advantages such as equipment manufacturing, smart terminals and smart sensors, and focus on promoting air, sewage, garbage and solid waste pollution prevention and control equipment, energy-saving equipment,

Instrument research and development and manufacturing, forming the core pillar of the local environmental protection industry. Connecting with the development of new materials industry, actively promoting the research and development and manufacturing of energy-saving materials, environmentally friendly materials and preparations, improving the environmental protection industry system, and driving the city

style transformation.

(III) Building a clean and efficient energy system

Strictly control the total amount of coal consumption. Scientifically arrange the new energy consumption demand during the "14th Five-Year Plan" and continue to implement coal consumption substitution. All reconstruction coal-consuming projects shall implement coal reduction substitution at 2 times or the highest provincial standard.

The total carbon consumption control has achieved the expected target set by Henan Province.

Accelerate the optimization and upgrading of energy structure. Promote the "introduction of heat into Zhengzhou" and "introduction of external electricity into Zhengzhou", advance the implementation plan of coal-fired power unit transformation and upgrading, and encourage the appropriate development of cogeneration units under the premise of technical and economic rationality.

The purpose is to support the development of clean energy such as pumped storage, wind power, photovoltaics, and hydrogen energy, and increase the utilization ratio of clean energy. Implement the "West-to-East Heat Transmission" policy, increase the construction of long-distance heat transmission networks, and alleviate the main urban area and Zhengzhou Airport Economic Zone.

Promote the construction of Kaifeng-Zhengzhou-Luoyang pipeline and improve natural gas supply capacity. Strengthen the construction of refined oil transportation and storage, improve storage facilities, and ensure stable oil supply. Build energy channel security protection

Energy channels operate safely and reliably. Implement innovative applications of energy big data, plan and build smart power stations, virtual power plants, and distributed energy station projects, and promote the intelligent development of all fields and links of energy.

External electricity accounts for 40% of the total electricity consumption in society, and the proportion of renewable energy and natural gas energy has increased significantly.

Improve energy efficiency. Strictly implement the "dual control" of total energy consumption and intensity, promote energy budget management and regional energy evaluation system, and expand the energy rights market to an annual comprehensive energy consumption of 5,000 tons of standard coal.

We will further promote energy conservation and efficiency improvement in key areas such as industry, transportation, construction, and public institutions, and strengthen the application of advanced energy-saving technologies, products, and equipment. We will strengthen the implementation of energy-saving review and supervision systems, and improve energy management.

Improve the regional performance evaluation system.

4. Building a green transportation system

Optimize and adjust the freight transportation structure. Promote the "road to rail" of bulk goods, actively promote the construction of dedicated railway lines, accelerate the "653 Project" of dedicated railway lines into enterprises and parks, and implement the main marshaling of railway trunk lines.

Expand capacity and build a collection and distribution system with interconnected and multimodal transport. For medium and long distance transportation of bulk goods such as coal and ore, railway transportation is the main method, while for short distance transportation, closed belt corridors or new energy vehicles are preferred.

Mining enterprises are encouraged to use National VI and above diesel trucks or new energy trucks for transportation. By 2025, the annual transportation volume of bulk goods such as coal, steel, electrolytic aluminum, electricity, cement, etc. will exceed 1.5 million tons.

Large logistics parks have basically achieved dedicated railway line connections.

Accelerate the promotion of green freight delivery in cities. Accelerate the construction of demonstration projects for green freight delivery in cities. New energy light logistics vehicles are mainly used for urban freight transportation, and new energy vehicles are preferred for short-distance freight between cities.

We will accelerate the construction of a green freight distribution demonstration project in Zhengzhou and speed up the construction of a provincial-level high-efficiency urban and rural distribution pilot project in Zhengzhou. We will establish a "external centralized transportation and internal distribution" urban green freight distribution system with centralized transportation by railways and internal distribution by new energy vehicles.

Import building materials, agricultural and sideline products, light industry and medicine and other production and living materials by road and rail transport.

Promote the use of new energy vehicles. Promote the use of new energy vehicles in the city's public sector. In addition to retaining some emergency vehicles and new energy vehicles that cannot meet the demand for use, add and update public buses and taxis.

All new energy vehicles will be used for public transportation (including public transportation and online taxis), sanitation vehicles, urban postal express, and urban logistics distribution vehicles; in principle, all new and updated official vehicles will be new energy vehicles.

Except for situations where ice and snow, refueling, emergency support and new energy vehicle technology cannot meet the requirements, the proportion of new and updated on-site vehicles to be electrified should reach 100% in principle. Accelerate the construction of charging facilities, highway service areas

50% of residential communities will have charging facilities, and the ratio of public charging piles to electric vehicles will be no less than 1:8. By the end of 2021, all urban buses and cruising taxis will be powered by new energy; by 2023

All trucks, cement tankers and sanitation vehicles have been updated to new energy vehicles.

Carry out the construction of low-emission driving areas. Designate low-emission driving areas in the city's built-up areas, and implement differentiated control of heavy pollution weather for transport trucks, diesel trucks and other vehicles entering low-emission driving areas.

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Column 3 Key projects for structural adjustment
1. Optimize and upgrade industries. Build vein industrial parks in Dengfeng, Gongyi, and Xingyang, and build 12 new green factories and 12 green 2 color industrial parks and 2 green supply chain management companies. 2. Adjust the transportation structure. Move the Zhengzhou Northeast Outer Passenger Transportation Loop Line to the north of the Yellow River and to the new Pingyuan Road in Xinxiang City. The district will study the planning of setting up Zhengzhou North Station. Build the China-Europe Express train assembly center project and Zhengzhou South Station high-speed rail express base Engineering. Construction of Zhengzhou CRRC Sifang Railway Vehicle Co., Ltd. dedicated railway, national refined oil reserve capacity construction 73 Seven engineering railway projects and a dedicated railway project of Guodian Xingyang Coal-Electricity Integration Co., Ltd. 3. Optimize the energy structure. Promote the integration of wind, solar, hydropower, and thermal energy storage, as well as the integration of source, grid, and load storage, and develop pumped storage power generation. Station construction and accelerate the construction of wind power projects.

5. Control greenhouse gas emissions and actively respond to climate change

1. Formulate an action plan to reach the peak of carbon emissions

Focusing on the national goal of reaching the peak of carbon emissions before 2030 and striving to achieve carbon neutrality before 2060, we will study and formulate an action plan for reaching the peak of carbon dioxide emissions in Zhengzhou, clarify the peak Ensure that carbon emissions peak before 2030. Carry out medium- and long-term carbon neutrality research and actively participate in national carbon emission rights trading.

Strengthen source control measures such as industrial structure and energy structure adjustment, actively develop low-carbon industries, low-carbon transportation, and low-carbon buildings, advocate low-carbon life, and focus on energy, industry, construction, transportation, and services. In key areas such as agriculture, we will formulate carbon emission peaking plans for key areas, promote key energy-consuming industries such as nonferrous metals and building materials to formulate carbon emission peaking work plans, and resolutely curb the blind development of "two high" projects. Improve coal The town's sewage treatment plant recycles and utilizes methane, and continuously increases the proportion of urban and rural domestic waste incineration treatment.

(ii) Enhance the ability to adapt to climate change

Improve agricultural adaptability to climate change and ensure food security. Improve urban and rural extreme climate event monitoring and early warning, disaster prevention and mitigation comprehensive assessment and risk management capabilities to ensure urban and rural construction and infrastructure safety. We will enhance the carbon sequestration capacity of natural ecosystems such as green spaces, conduct risk assessments of climate change impacts when appropriate, and implement actions to adapt to climate change.

3. Promoting the integration of climate change management

Carry out carbon monitoring pilot projects and gradually incorporate greenhouse gas monitoring into the ecological environment monitoring system for overall implementation. Promote the integration of climate change response and environmental statistics, environmental assessment, environmental monitoring, environmental law enforcement and other environmental Improve the greenhouse gas inventory compilation and dynamic update work mechanism, incorporate it into environmental statistics, and gradually improve the greenhouse gas emission statistical accounting system. Incorporate climate change response requirements into the "three lines and one list" ecological Through planning and project environmental assessments, we will promote regions, industries and enterprises to implement policy requirements such as coal consumption reduction and substitution, greenhouse gas emission control, and incorporate climate change impacts into environmental impact assessments. We will establish a system for publicizing information on environmental protection, and uniformly organize and implement ecological and environmental supervision and law enforcement in accordance with laws and regulations. We will strengthen the responsibility system for controlling greenhouse gas emissions targets, and increase the assessment of work on addressing climate change.

4. Carry out low-carbon pilot demonstration projects

Explore and carry out pilot demonstrations of near-zero carbon emissions and carbon neutrality. Implement the "dual-target" coordinated control of air pollutants and greenhouse gas emissions, and create low-carbon parks and low-carbon communities. Carry out electricity, building materials, etc. Carry out "carbon labeling" and "carbon neutrality" practices, and encourage the development, demonstration and promotion of technologies related to the coordinated reduction of greenhouse gases and pollutants.

(V) Cultivating a green and low-carbon lifestyle

Raise the public's awareness of green and low-carbon. Strengthen publicity and education, and encourage the public to start from themselves, choose a simple, moderate, green and low-carbon lifestyle, and oppose extravagant waste and unreasonable consumption. Carry out the creation of festivals Through actions such as green families, green schools, and green communities, we can improve the ecological and cultural literacy of the entire population and make the low-carbon and environmentally friendly green lifestyle deeply rooted in the hearts of the people.

Advocate for green product consumption. Promote the government procurement system for green products, combine it with product list management, and increase the application of green product-related standards in government procurement. State-owned enterprises should take the lead in implementing the Encourage other enterprises to carry out green procurement on their own initiative. Actively play the leading role of green consumption and vigorously promote energy-saving, environmentally friendly and low-carbon products. Resolutely stop food waste and actively practice the "Clean Plate Campaign". Plastic bags, and strengthen supervision and inspection of the implementation of the "plastic restriction order".

Vigorously promote green and low-carbon travel. Build a transportation model that combines public transportation with bicycle transfers and park-and-ride transfers, and build a complete, convenient, safe and transfer-friendly bicycle and sidewalk system. Participate in green practices and develop a green and low-carbon travel style.

We will further promote the development of green buildings. We will continue to improve the energy efficiency of buildings and promote green building design standards. We will improve the building energy consumption quota management system, comprehensively promote the use of renewable energy in new buildings, and continue to Energy efficiency, carry out demonstration construction of ultra-low energy consumption buildings. By 2025, the proportion of newly built green buildings will reach 100%.

VI. Strengthen coordinated control and improve air quality

1. Promote in-depth pollution control in industrial enterprises

Promote performance grading management in key industries. Standardize and strengthen performance grading management of key industry enterprises, improve the evaluation mechanism, and implement dynamic performance grading management. Cultivate and promote enterprises to "gradiently meet standards", We will strive to upgrade the overall governance level. We will implement relevant incentive policies for Class A and Class B enterprises and play a leading role in advanced demonstration.

Carry out waste gas treatment for cluster enterprises in the park. Promote the upgrading and transformation of professional parks such as building materials, aluminum processing, casting, and refractory materials in the construction of "parks in the park", and formulate comprehensive remediation plans of "one park, one policy" and "one industry, one policy". The new projects of the Bureau will be strictly approved and checked to prevent pollution from spreading to the countryside. Open-air and open-type spraying operations will be completely banned, and the construction of centralized spraying engineering centers in the park will be promoted.

Consolidate the results of industrial pollution control. Consolidate the ultra-low emission transformation results of the steel and cement industries, and implement differentiated electricity and water price policies as required. Further strengthen the denitrification technology upgrade in the thermal power industry, and strengthen Operation and management of desulfurization and denitrification facilities, effective control of ammonia escape, strengthening the whole process control of pollutant emissions, and collaboratively reducing the total amount of pollutant emissions by participating in carbon emission rights market transactions. Install online monitoring equipment for ammonia escape in desulfurization and other treatment devices, and connect them to the ecological and environmental departments; achieve ultra-low emissions for all industrial furnaces by the end of 2025.

(II) Reducing emissions of volatile organic pollutants

Optimize the structure of raw and auxiliary materials and products containing VOCs. Strictly control the production and use of high-VOCs content solvent-based coatings, inks, adhesives, cleaning agents and other construction projects. In principle, no new construction will be carried out.

Production enterprises should speed up product upgrading and transformation, and increase the proportion of water-based, high-solid, solvent-free, powder and other low-VOCs products. Industrial coating, packaging and printing, electronics and other industries should formulate work plans.

By the end of 2025, the source substitution capacity of raw and auxiliary materials with low VOCs content will be reduced by 10%.

All outdoor structure protection and road traffic signs use low-VOCs paint. Strictly implement the VOCs content limit standards for paints, inks, adhesives, and cleaning agents, increase random inspections, and ensure production

By 2025, the use of solvent-based industrial coatings and solvent-based inks will be reduced by 20% and 15% respectively, and the use of solvent-based adhesives will be reduced by 20%.

Strengthen comprehensive management of VOCs in all links. Further strictly investigate five types of emission sources, including storage, transfer and transportation of VOCs-containing materials, leakage of equipment and pipeline components, escape from open liquid surfaces, and process.

Take measures such as sealing equipment and places, improving processes, and effectively collecting waste gas to comprehensively manage VOCs unorganized emissions. In accordance with the principle of "collecting all that should be collected and collecting by quality", transform unorganized emissions into

Centralized treatment, selection of appropriate and efficient treatment technology, strengthening operation and maintenance management, treatment facilities should be "started before stopping" compared with production equipment, and the waste gas collection rate, treatment facility synchronous operation rate and pollution control rate should be improved.

VOCs are stably discharged up to standard. During the start-up, shutdown, inspection and maintenance of the enterprise, VOCs waste gas generated by operations such as material return, cleaning and purging shall be collected and treated in a timely manner as required.

3. Further efforts to prevent and control diesel truck pollution

Actively promote the elimination of old vehicles. Fully implement the National 6b emission standards for light and heavy vehicles, and vigorously promote the early elimination and renewal of old motor vehicles. In accordance with the work requirements of Henan Province, fully eliminate the National III and above

Oil and gas trucks (including on-site operation vehicles), complete the elimination task of diesel trucks that meet the National IV emission standards and gas trucks that use lean burn technology issued by Henan Province;

The industry shall register, dismantle and destroy the waste in accordance with the regulations.

Strictly supervise motor vehicles to ensure they meet environmental standards. Strengthen the construction and management of road inspection and law enforcement inspection sites, complete the standardization of road inspection and road inspection points at national and provincial entrances, increase road inspections of heavy trucks and centralized

Conduct door-to-door inspections and crack down on illegal activities such as removing exhaust gas aftertreatment devices, damaging and tampering with the on-board diagnostic system (OBD). Fully implement the vehicle emission inspection and maintenance (I/M) system and vehicle emission recall

Oil evaporation emission control testing. Establish a comprehensive access control system for bulk logistics and transportation enterprises, and establish an electronic ledger for transportation vehicles, in-factory vehicles, and non-road mobile machinery. Implement the "Henan Electronic Pass Integrated Platform"

Strengthen traffic supervision of vehicles with excessive emissions.

Strengthen the comprehensive management of non-road mobile sources. Fully implement the fourth stage emission standards for non-road mobile diesel machinery. Accelerate the upgrading and transformation of new energy vehicles and machinery for internal operation of factories and mining enterprises and units.

We will carry out special rectification actions for vehicles (machines) within the enterprise, conduct surveys of mines in the city, collect information and conduct emission tests on vehicles (machines) in mines and enterprises, and promote the replacement of high-emission vehicles (machines) with new energy.

By the end of the year, non-road mobile machinery with national emission standards of or below will be basically eliminated. We will strengthen the supervision of non-road mobile machinery manufacturers and the management of emission control areas, accelerate the collection of non-road mobile machinery information, and comprehensively

Black smoke is emitted from machinery and railway diesel locomotives.

Strengthen the supervision of oil product quality. All districts, counties and cities should conduct regular inspections and on-site spot checks on gasoline (including alcohol-containing gasoline, aviation gasoline), aviation kerosene and other commercial oil storage within their jurisdiction.

Supervision and inspection: conduct supervisory sampling of oil quality at more than 5% of gasoline stations in operation. Increase the inspection of the installation and use of oil and gas recovery facilities in the entire process of oil storage, transportation and sales, and strengthen the oil and gas recovery facilities.

Supervision and management of automatic monitoring equipment for oil and gas recovery at gas stations with a gasoline sales volume of more than 5,000 tons. Resolutely crack down on non-standard oil products and strengthen all aspects of diesel import, production, storage, sales, transportation, and use.

4. Strengthen comprehensive control of surface-source air pollution

Strengthen comprehensive dust control. Carry out intelligent construction of construction sites, and strictly implement the "eight hundred percent" and "two prohibitions" (prohibiting on-site mixing of concrete and on-site preparation of mortar) measures.

Enter the scope of daily safety and civilized construction supervision, organize the early warning of heavy pollution weather, control dust on construction sites and roads under windy weather conditions, and establish a reporting supervision and open and secret investigation work mechanism. Large coal

The material yards have completed the construction of dust suppression facilities such as fencing, covering, and automatic spraying, and the closed transformation of material conveying systems. We encourage yards with conditions to implement full-closed transformation. Further expand mechanized road cleaning

Improve the cleaning capacity of important sections such as outer ring roads of large cities, urban entrances and exits, and urban-rural junctions to improve the cleaning quality of urban roads. By 2025, the mechanized cleaning rate of roads in urban built-up areas of districts, counties (cities) will be

The average dust fall in the area shall not exceed 6 tons/month/square kilometer, and grid-based dust fall monitoring and assessment shall be implemented.

Continue to promote green mine construction. Strictly implement green mine construction standards, determine green mine construction goals, and implement green mine incentive policies. Promote the comprehensive development of mineral mining and sand and gravel enterprises.

To deal with the unorganized emission pollution problems existing in various production links such as raw material transportation, storage, loading and unloading, crushing, transfer, screening, and discharging, we will control, collect, purify and treat the whole process.

And corresponding pollutant emission monitoring equipment, all large and medium-sized open-pit mines in production in the city have been turned into green mines.

Carry out the pollution control of restaurant fume and odor. Strengthen the source control of fume nuisance, promote the implementation of third-party operation and maintenance management and operation status monitoring of fume treatment facilities. Strengthen the management of sewage treatment, garbage disposal, livestock

Prevention and control of odor pollution in industries such as material products.

Control the emission of large-scale air pollutants from agricultural sources. Encourage the closed management of large-scale pig and chicken pens, promote the closure of manure transportation, storage and treatment facilities, and strengthen waste gas collection and treatment. By 2025,

Total atmospheric ammonia emissions reduced by 5%.

Further expand the scope of the no-burning zone. Five districts in the city (including Jinshui District, Erqi District, Zhongyuan District, Guancheng Hui District, Huiji District) and four development zones (including Zhengzhou Airport Economic Zone, Zhengdong New District, Zhengzhou

The entire administrative area of the High-tech Zone is designated as a high-pollution fuel ban zone, and the Shangjie District and counties (cities) will further expand the high-pollution fuel ban zone based on the completion of clean heating transformation.

(V) Make every effort to eliminate heavy pollution

Significantly reduce the frequency of heavily polluted weather. Make full use of the national-provincial-municipal three-level prediction and forecasting technical support of the Ministry of Ecology and Environment, and conduct joint consultations with the meteorological department and expert teams. Continue to carry out autumn and winter

Immediately improve the dynamic update mechanism of the emergency emission reduction list for heavy pollution weather, scientifically and accurately classify and implement emergency emission reduction measures for heavy pollution weather in specific areas for industrial enterprises with different governance levels and emission intensities.

By 2025, the proportion of days with severe pollution or above in the city will be no more than 1.4%.

Strengthen the coordinated control of O3 and PM2.5 in summer. Strengthen the ability to accurately forecast O3 in summer, improve the coordinated control response mechanism of O3 and PM2.5, formulate an action plan to tackle O3 pollution in summer, and gradually expand the coordination of O3 and PM2.5.

Guide relevant units and enterprises to formulate "one factory, one policy" implementation plans and take effective measures to reduce pollutant emissions; consider introducing incentive policies in due course to reduce gas stations, oil storage depots, and key VOCs enterprises.

Pollution emission intensity.

Strengthen regional joint prevention and control and law enforcement inspections. Strengthen joint prevention and control with surrounding cities to reduce the intensity of pollutant emissions in the entire region and minimize the impact of transmission. Law enforcement should be strengthened in heavy pollution weather.

Division of responsibilities, make full use of drones, aerial inspections and other methods, and conduct "double random, one public" inspections on the operation of air pollution prevention and control facilities, the consumption of raw and auxiliary materials, the disposal of hazardous waste, etc.

On-site inspection, check raw and auxiliary material test reports, ledgers, operation records, purchase invoices, hazardous waste disposal forms and other records.

Column 4 Key projects for air pollution control
1. Comprehensive control of VOCs. Xingyang Valve Industrial Park 5G Intelligent Shared Spraying Center Project, Xingyang City Five Long Industrial Park 5G Intelligent Shared Spraying Center Project, Zhengzhou Hongjing Color Printing and Packaging Co., Ltd. VOCs Emission Deep Treatment Zhengzhou Century Jingxin Machinery Manufacturing Co., Ltd. Injection Molding VOCs waste gas treatment project, Zhengzhou Yunhao Auto Parts Co., Ltd. automatic spraying section organic waste gas treatment facility upgrade Level transformation project. 2. In-depth governance of industrial enterprises. Build a centralized solvent recovery and disposal center in the automobile industrial park of Zhengzhou Economic Development Zone, A centralized recycling center for waste activated carbon is set up, and Zheng Coal Comprehensive Machinery has an annual output of 4,400 hydraulic brackets for air pollution prevention and control projects. 3. Ammonia emission control projects. Large-scale ammonia emission reduction projects in large-scale farms, electricity, cement, steel, etc. Key industries will improve denitrification engineering facilities to reduce ammonia escape.

7. Deepen the coordination of three waters and steadily improve the water ecological environment

1. Improving the level of drinking water safety

Pay attention to the water source protection of the middle route of the South-to-North Water Diversion Project. Carry out in-depth investigation and rectification of environmental problems within the protection area of the middle route main canal (Zhengzhou section) of the South-to-North Water Diversion Project, and continuously consolidate the rectification results. Strengthen regular monitoring. Improve the water quality of the Central Line Main Canal (Zhengzhou Section). Comprehensively improve the standardized construction level of the protected area, effectively prevent water environmental risks, effectively eliminate environmental risk hazards, and improve the level of water quality safety assurance. The middle route main canal of the South-to-North Water Diversion Project (Zhengzhou section) has been incorporated into the management system of the river and lake chief system, and daily management is carried out in accordance with the responsibilities of the river and lake chief system.

Continue to promote the standardized construction of urban drinking water sources. Deepen the standardized construction of centralized drinking water sources in cities at the county level and above, regularly inspect and update damaged boundary markers and signs in protected areas. etc., strengthen isolation and protection construction, and strengthen risk prevention and control and emergency response capacity building. Focus on promoting the adjusted Huanghe Mangshan and Huayuankou surface water source protection areas and Jiuwutan underground water source protection area. Accelerate the installation of video surveillance equipment at county-level centralized drinking water sources, and strengthen the early warning and monitoring capabilities of surface water drinking water sources. Carry out the "return to the source" campaign to rectify environmental problems in drinking water source protection areas. We will rectify each place and implement dynamic "zero clearance" to strictly prevent the resurgence of the problem and ensure drinking water safety.

Steadily promote the protection of rural drinking water sources. Accelerate the demarcation of rural centralized drinking water source protection areas (scopes), conduct inspections of rural drinking water source protection areas throughout the city, and regularly conduct Assessments will be conducted to ensure that water quality is stable and up to standard. By 2023, the demarcation and demarcation of township-level centralized drinking water source protection areas will be completed. Opening force.

(II) Continue to deepen water pollution control

Accelerate the construction and renovation of urban sewage treatment facilities. According to the principle of moderately advanced urban sewage treatment capacity, improve the scale and overall layout of urban sewage treatment facilities. The treatment plant should carry out upgrading and transformation as soon as possible to ensure that the effluent can stably meet the requirements of the water pollutant discharge standards of the Yellow River Basin in Henan Province; according to the utilization of recycled water and the needs of urban water ecology, promote the differentiated precision of sewage treatment facilities. Sewage treatment plants should be upgraded and renovated to meet surface water quality standards of Class V or above before discharge. Sewage treatment plants at or above the county level that meet the conditions should build tailwater artificial wetlands.

Comprehensively promote pollution prevention and control in industrial parks. Improve sewage and garbage collection and centralized treatment facilities in industrial parks, and promote third-party governance of environmental pollution in industrial parks. Strictly control untreated or ineffectively treated industrial wastewater. Water treatment system, severely crack down on illegal discharge and direct discharge.

Implement the rectification of sewage outlets into rivers. Focus on the Yellow River main stream, Yiluo River, Sishui River, Jialu River, Shuangli River, Ying River and other rivers, conduct a comprehensive and in-depth investigation of sewage outlets into rivers, and establish an information platform for sewage outlets into rivers. In accordance with the requirements of "recruiting one batch, merging one batch, and standardizing one batch", we will implement classified remediation of sewage outlets into rivers. We will establish a sewage outlet remediation and cancellation system, form a list of sewage outlets that need to be retained, and carry out daily supervision and management. The city will conduct water function zoning adjustment and revision work in due course. By 2025, the sewage outlets of important rivers will be rectified, and a long-term mechanism with standardized management and effective supervision will be formed.

Continue to carry out the treatment of black and odorous water bodies in built-up areas. Give full play to the long-term role of rivers and lakes, consolidate and improve the treatment results of black and odorous water bodies in built-up areas of the city, and establish a long-term mechanism to prevent the return of black and odorous water bodies. All counties (cities) have completed The black and odorous water bodies will be investigated and remediation plans will be formulated, and a unified list of black and odorous water bodies and deadlines for reaching the standards will be announced. By 2025, black and odorous water bodies will be basically eliminated in county-level urban built-up areas.

3. Ensuring the ecological flow of rivers and lakes

Build a regional circulating water system. Continue to promote the ecological water system circulation project around Zhengzhou City to realize the recycling of ecological water in the urban area. Accelerate the implementation of the Putianze water circulation project in Zhengzhou City to recycle water from the lower reaches of the Jialu River. The water of Fenghe River, Changqu River, Chaohe River, Shibali River, Shiqili River, Xiong'er River and Jinshui River in the urban area is diverted to increase the ecological flow of urban rivers.

Strengthen the ecological water security of rivers and lakes. Gradually reduce the intensity of water resource development and utilization in over-exploited rivers and regions, and reduce the ecological water that has been squeezed out. Establish and improve the ecological water security mechanism, and classify and categorize the ecological water security mechanism. We will set the goal of ensuring water quality, formulate ecological flow guarantee plans for important rivers and lakes such as the Jialu River, Shuangli River, and Ying River, and strengthen ecological flow monitoring, evaluation, and early warning at major control sections of important rivers and lakes.

(IV) Strengthening water ecological protection and restoration

Promote ecological management of rivers and lakes. Strengthen the construction of water ecological protection and restoration projects, and improve river ecology through measures such as river ecological buffer zone construction, aquatic biodiversity restoration, and precise upgrading of sewage treatment plants. Pilot ecological transformation of non-flood drainage river channels with revetments or "three-sided" hardening will be implemented to restore the natural purification function of the coastline and water bodies.

Promote the cleanup and rectification of river systems. Carry out special rectification actions such as "clearing four chaos" and "three pollutions and one clean" in rivers and lakes, and conduct rectification of direct discharge outlets (ditches) into rivers that directly affect the water quality of rivers and lakes, water-related discharge enterprises, and "scattered We will investigate and rectify farms, restaurants, dry toilets, garbage (straw), and fertilization and planting of crops on the banks, slopes and beaches. We will continue to deepen the implementation of the river and lake chief system and improve the "river chief + police chief, river chief + prosecutor, Protection mechanisms will be established to enhance the supervision and protection capabilities of rivers and lakes.

Strengthen ecological restoration after severe rainstorms. Strengthen comprehensive ecological environment improvement around the South-to-North Water Diversion Middle Route Main Canal (Zhengzhou Section) to eliminate pollution and secondary ecological risks. Restore the water conservation capacity of the affected land. Improve flood control and retention capacity. Repair damaged river wetlands and related facilities.

Carry out water ecological health assessment. Improve the protection of sensitive areas such as Zhengzhou Yellow River Wetland Provincial Nature Reserve, Zhengzhou Yellow River National Wetland Park, and Yellow River Zhengzhou Section Yellow River Carp National Aquatic Germplasm Resources Protection Area.

The monitoring work was carried out to optimize monitoring points, improve monitoring factors, and evaluate the water ecological health status of the Jialu River in Zhengzhou.

Actively promote the construction of beautiful rivers and lakes. Focusing on the Jialu River, we will actively carry out the construction of beautiful rivers and lakes, promote governance through construction, and strive to create a number of rivers and lakes with smooth rivers, clear lakes, green banks and beautiful scenery, and realize the goal of "clear water, green

By 2025, we will strive to build the Jialu River into a beautiful river and lake with national demonstration value, and form good experience and practices in the protection and construction of beautiful rivers and lakes.

Column 5 Key projects to improve the water ecological environment
1. Construction and improvement of sewage treatment facilities. Phase II of Zhengzhou New District Sewage Treatment Plant and Phase I of Nancao Sewage Treatment Plant Engineering, Xingyang New Materials Park Wastewater Treatment Plant, Shangjie District Third Wastewater Treatment Plant, Gongyi Private Science and Technology Park Wastewater Treatment Plant Treatment plant, Xinzheng Xingang Industrial Cluster Zone Electronic Information Industry Park supporting sewage treatment and reuse project. Zhengzhou Shangjie No.1 Sewage Treatment Plant, Zhengzhou Shangjie No.2 Sewage Treatment Plant, Gongyi Beijing Enterprises Water Services Co., Ltd. (No. Gongyi City Beijing Enterprises Water Services Co., Ltd. (Phase II), Guangwu Town Wastewater Treatment Plant, Xingyang Jiangshuo Water Services Co., Ltd. The company has carried out upgrading and renovation, and the effluent has reached or exceeded the Yellow River Basin Water Pollutant Discharge Standard (DB41/2087- 2021ÿÿ 2. Water system connection project. Implement the Putianze water circulation project in Zhengzhou. 3. Ecological restoration projects for key rivers and lakes. Zhengzhou Yellow River Beach Park construction project, Shuangli River, Dengfeng Shaoyang The comprehensive management project of the Hehe Shuyuan River.

8. Promote systematic prevention and control to ensure soil and groundwater safety

1. Strengthening the prevention and control of soil pollution sources

Strictly control the spatial distribution of soil pollution. Incorporate soil and groundwater environmental requirements into national land space planning, rationally plan land use based on soil pollution conditions and risks, and strengthen the implementation of soil pollution risk control.

"One map" management of information and national spatial planning. Ensure environmental access to construction projects and strictly control the implementation of projects involving heavy metals and those that do not meet environmental control requirements.

Strengthen the prevention and control of soil pollution sources. Combined with the results of land use surveys of key industries and enterprises, dynamically update the list of key soil pollution supervision units, specify soil pollution prevention and control requirements in the pollutant discharge permit, and encourage

Continue to carry out soil pollution surveys in typical industries and their surrounding areas, agricultural land in areas with high heavy metal geological backgrounds, etc., to further understand the soil pollution situation. Urge key soil pollution supervision units to carry out soil pollution investigations.

Continue to promote the investigation and rectification of cadmium-related heavy metal enterprises around cultivated land, conduct investigation and analysis of the causes of soil pollution in contaminated cultivated land, propose targeted source-cutting measures and implement them as a priority, and carry out orderly

Pollution source prevention and control actions. Focusing on counties (cities) with concentrated mineral resource development activities such as Dengfeng, Gongyi, and Xinmi, we will comprehensively apply remote sensing, drones, and on-site surveys to comprehensively investigate disorderly stacking of wastewater in mining areas.

Explore the construction of national-level soil pollution prevention and control pilot areas.

Strengthen the supervision of key supervision units. Improve the soil monitoring system around key soil pollution supervision units, and regularly conduct soil environmental monitoring around key soil pollution supervision units. Gradually promote the supervision of key soil pollution supervision units such as cadmium.

The list of key air and water pollutant dischargers will be added to the list, and automatic monitoring will be implemented for the emission of atmospheric particulate matter and the emission of heavy metals such as cadmium in wastewater. By 2025, enterprises in the industry involving heavy metals such as cadmium will be included in the list of key air and water pollutant dischargers.

The automatic monitoring equipment for air and water pollutant emissions is installed and connected to the network for use.

(II) Ensuring safe use of cultivated land and construction land

Consolidate and improve the classified management and safe use of agricultural land. Strictly protect priority protected agricultural land to ensure that its area does not decrease and the soil environmental quality does not decline. Implement the "one map and one table" for safe use of cultivated land.

Responsibility management. Summarize the effectiveness of existing safe utilization measures, formulate a work plan for safe utilization of cultivated land and promote its implementation in an all-round way. Strengthen the supervision of strictly controlled cultivated land and demarcate specific agricultural products strictly controlled areas in accordance with the law.

Agricultural products; encourage measures such as adjusting planting structure and returning farmland to forest and grassland to ensure safe use of strictly controlled farmland. Dynamically adjust the

We will classify and sort agricultural products according to their quantity, and gradually improve the archive information of cultivated land soil environment quality. We will formulate emergency disposal plans for agricultural products that exceed the standards, and strictly prohibit agricultural products that exceed the standards from entering the circulation market.

Implement risk control and governance and restoration of construction land in an orderly manner. Continuously update the risk control and restoration list of construction land to achieve the management of "one negative list" for development and utilization. Conduct soil pollution surveys in accordance with the law.

Priority will be given to conducting further investigations and risk assessments of potentially high-risk plots identified through land use surveys of key industries and enterprises. Strictly control the planned use of heavily polluted plots in industries such as pesticides and chemicals, and strengthen pollution prevention and control.

Prevent illegal development and utilization. Formulate differentiated measures to manage risks of contaminated land that is not being developed and utilized. Focus on contaminated land that has been changed to residential, public management and public service land, and make full use of the land.

We will implement soil pollution control and restoration for key construction sites. We will promote the concept of green restoration and strengthen the prevention and control of secondary pollution during the restoration process. Where there is groundwater pollution, we will coordinate the promotion of soil and groundwater risk management.

Strengthen the joint supervision of soil environment of construction land. Strictly manage the access to construction land, improve the management policy of access land, and strengthen the joint supervision mechanism among departments. Xingyang City and Gongyi City have carried out district, county (city) level

The two departments of resource planning and construction land joint supervision pilot project will explore the establishment of institutional mechanisms in information sharing, land access, planning comparison, etc. For risk control plots, through tracking monitoring and on-site inspections, etc.

Improve the post-process management of risk control and restoration activities.

3. Implementing groundwater pollution risk control

Promote the zoning management of groundwater environment. According to the results of the "dual source" survey, complete the demarcation of groundwater pollution prevention and control zones in the whole city, scientifically demarcate groundwater pollution prevention and control zones, and preliminarily determine the protection zone and protection zone.

Scope and zoning prevention and control measures, explore the establishment of a list of key groundwater supervision units. Do a good job in the construction of "groundwater pollution prevention and control pilot areas". Gradually complete the centralized groundwater drinking water source at the county level and above

Establish an inventory of pollution sources in the recharge area, assess pollution risks, and develop and implement strong protection measures.

Strengthen groundwater quality target management. Focus on the national groundwater assessment points in the "14th Five-Year Plan", analyze the water quality status, implement water quality consolidation or improvement actions according to needs, and complete the national groundwater quality assessment.

The ecological and environmental departments of the districts, counties (cities) where the national groundwater test sites are located will strengthen groundwater environmental supervision to ensure that the water quality is stable and meets the standards.

Continue to carry out groundwater environmental status surveys and assessments. Focusing on the "dual sources", carry out groundwater environmental status surveys in groundwater drinking water source protection areas, recharge areas, and areas surrounding water supply units.

Water source safety; continue to conduct surveys and assessments of groundwater pollution sources and surrounding areas such as chemical production enterprises, industrial clusters, tailings ponds, hazardous waste disposal sites, landfills, and mining areas.

Before that, in accordance with national and provincial requirements, a number of groundwater environmental status surveys and assessments of other pollution sources were completed.

Promote the prevention of groundwater pollution at the source. Focus on areas such as the Yellow River Basin and the South-to-North Water Diversion Middle Route Main Canal (Zhengzhou Section) to strengthen groundwater pollution risk management. Promote chemical production enterprises,

Key enterprises such as landfills and landfills should implement anti-seepage measures and carry out anti-seepage transformation. Accelerate the construction of leachate treatment facilities at landfills and strictly manage them on a daily basis.

Promote groundwater pollution risk control by category. Develop groundwater pollution risk control plans for key pollution sources such as industrial clusters, hazardous waste landfills, and garbage landfills, and conduct risk management and control for areas with potential risks.

Risk control measures are needed. The list of abandoned mines, drilling wells, and water wells should be continuously improved and updated dynamically, and well sealing and backfilling work should be continuously carried out.

4. Deepen the remediation of agricultural non-point source environment

Strengthen the prevention and control of non-point source pollution in the planting industry. Improve the survey and statistical system for the use of fertilizers and pesticides, strengthen the standardized management of agricultural inputs, improve the input traceability system, and promote the green development of agriculture.

We will carry out actions to further promote soil surveying and formula fertilization, replace chemical fertilizers with organic fertilizers, explore new ways to organically combine it with the return of livestock and poultry manure to the fields, and promote the application of new pesticide green prevention and control technology and high-efficiency large and medium-sized plant protection machinery.

Yuan, carry out non-point source pollution load assessment, identify key regulatory areas and important time periods, and compile a list of priority governance areas; select Gongyi City as a pilot to explore the assessment of agricultural non-point source pollution. By 2025

The application of fertilizers and pesticides maintained negative growth, and the utilization rates of fertilizers and pesticides for major crops reached more than 43%.

Strengthen pollution control in the livestock and poultry breeding industry. Strictly implement the regulations on livestock and poultry breeding prohibited areas, encourage the allocation of livestock according to the land area in suitable breeding areas, and promote the green development of animal husbandry. Accelerate the process of returning manure to the fields nearby and utilize the "last hectare" of land.

We will strengthen the prevention and control of livestock and poultry pollution in the lower reaches of the Yangtze River, promote the construction of centralized manure treatment centers, and coordinate the establishment of a network system and market-oriented operation mechanism for the collection, transformation and utilization of manure in the aquaculture industry.

Increase the density of aquaculture and strictly manage aquaculture inputs. Promote ecological and healthy aquaculture models and accelerate water conservation and emission reduction in aquaculture. Conduct a survey on the basic information and environmental status of aquaculture enterprises and standardize the management of aquaculture enterprises.

By 2025, all large-scale farms will be fully equipped with manure treatment facilities, and the comprehensive utilization rate of livestock and poultry manure in the city will reach more than 94%.

Coordinate and promote the resource utilization of agricultural waste. Strengthen the prevention and control of agricultural film sources, promote standard mulch films and biodegradable mulch films, and promote the reduction of mulch film sources. Strengthen the supervision of the use and recycling of agricultural films.

Establish a green compensation system for the recycling of waste agricultural films and pesticide packaging. Continue to carry out pilot projects for the recycling of waste agricultural films and pesticide packaging in Zhongmu County. Based on the experience of the pilot projects, accelerate the promotion of the recycling of waste agricultural films and pesticide packaging.

Treatment system construction: By 2025, the city will basically realize the recycling and treatment of all agricultural films.

Column 6 Key projects for soil, groundwater and agricultural pollution control
1. Soil environment status survey. In the vegetable and fruit production areas such as Zhongmu and Xingyang, as well as the grain production functional areas, In areas where heavy metal pollution of cultivated land is a prominent problem, long-term observation and research bases will be established in phases and batches to carry out soil pollution, Monitoring of heavy metals in atmospheric precipitation, irrigation water, fertilizers and other agricultural inputs. 2. Soil pollution control and remediation. Conduct pollution survey and risk assessment of sulfuric acid production sites in Zhengzhou. Soil pollution investigation and risk assessment of Henan Cape Chemical Co., Ltd., decommissioning of Zhengzhou Rambo Technology Co., Ltd. Polluted land remediation and restoration project in the factory area. 3. Groundwater pollution prevention and control. Investigation, assessment and sealing and backfilling of abandoned coal mines in the Chaohua section of Shuangli River Basin in Xinmi City Engineering, investigation and assessment of abandoned water wells in Zhengzhou Airport Economic Zone and well sealing and backfilling engineering, underground water environment engineering in the central urban area of Zhengzhou Environmental quality survey. 4. Agricultural non-point source pollution prevention and control projects. Agricultural film recycling project in Zhongmu County, pesticide packaging project in Zhongmu County and Huiji District Demonstration project for waste recycling.

IX. Coordinate urban and rural comprehensive management and improve urban and rural living environment

1. Improve environmental protection infrastructure construction

Improve the city's centralized heating capacity. Continue to accelerate the construction of heating pipe networks, fully release the heating capacity of cogeneration, industrial heating, etc., and eliminate coal-fired boilers and loose coal within the coverage of the pipe network.

We will build facilities, tap the potential of heating, promote the extension of abundant heating capacity to a reasonable radius, and accelerate the transformation of old urban centralized heating pipes and networks. By 2025, the penetration rate of centralized heating in Zhengzhou's urban built-up areas will strive to reach

The penetration rate of centralized heating in the built-up areas of the (city) has reached over 85%.

Comprehensively promote the renovation and construction of sewage pipe networks. Comprehensively promote the rain and sewage diversion renovation, mixed and wrong connection renovation, and damaged pipe network repair in urban areas and built-up areas of counties (cities), and eliminate the direct flow of domestic sewage in built-up areas.

For urban sewage treatment plants with an annual average oxygen demand concentration below 100 mg/L, a systematic remediation plan of "one plant, one policy" should be formulated around the service area network, with clear remediation goals and measures to improve the collection and utilization of the sewage network.

Water planning will be implemented, and the sewage collection network system in urban villages, old urban areas, and urban-rural junctions will be improved as soon as possible to eliminate the sewage network service blank areas. By 2025, 260 kilometers of new sewage network will be added.

The centralized collection rate of domestic sewage is greater than 90%, and the centralized collection rate of domestic sewage in towns and cities in all counties (cities) has been increased to 70%.

Promote rural domestic sewage treatment in accordance with local conditions. Accelerate the promotion of rural sewage treatment, and promote rural sewage treatment in accordance with the rural sewage treatment plans of various counties (cities) in accordance with local conditions. Improve the operation and maintenance of facilities

Ensure the stable and normal operation of rural sewage treatment facilities, and continue to promote full coverage of effluent monitoring of treatment facilities with a daily treatment capacity of 20 tons or more. Support Xinmi City and Xingyang City to carry out long-term operation and maintenance pilot projects for rural domestic sewage treatment

Supporting operation and maintenance management mechanism. By 2025, the rural domestic sewage treatment rate will reach 70%.

Promote the harmless resource-based treatment and disposal of sludge. Accelerate the harmless treatment and disposal of sludge in urban sewage treatment plants in accordance with the requirements of reduction, stabilization, harmlessness and resource utilization of urban sludge treatment and disposal.

Reduce the scale of sludge landfill, promote centralized sludge incineration and harmless treatment, and gradually establish a sludge treatment model with "biomass utilization + incineration" as the main method and power plants and cement kilns as the supplementary method. By 2025

The treatment and disposal rate reached 100%.

(II) Improving the construction of beautiful villages

Comprehensively improve the rural living environment. Implement the "Thousand Villages Demonstration, Ten Thousand Villages Renovation" project for rural living environment improvement, and further promote the construction of beautiful environment, beautiful fields, beautiful villages, and beautiful courtyards. Implement rural garbage,

Improve the rural domestic waste collection, transportation and disposal system, promote the classification and reduction of rural domestic waste, and actively promote the "two-four-division method" and other methods that are in line with rural characteristics and simple and environmentally friendly.

Improve resource recycling networks, accelerate the construction of rural domestic waste recycling systems, and carry out special actions to standardize the management of domestic waste landfills. Implement rural river network connection projects to build

Microcirculation, promote comprehensive management of rural rivers, and build ecological river channels and ponds. Promote rural toilet reform according to local conditions, properly handle sewage after toilet reform, and effectively control feces and sewage generated after toilet reform. By 2025

The classification and resource utilization have been basically realized; 500 beautiful rural demonstration villages and 100 high-quality villages have been built, and 357 new rural environmental comprehensive improvement tasks have been completed.

Strengthen the management and supervision of rural black and odorous water bodies. According to the different types of rural black and odorous water bodies, different management methods are adopted to eliminate external pollution and achieve "dynamic zero" of rural black and odorous water bodies in the whole city.

On the basis of investigation and verification, the basic information of rural black and odorous water bodies will be made public to the society through the websites of district, county (city) governments, and announced to the villagers through bulletin boards based on administrative villages. The supervision list will be dynamically updated.

We will strengthen local responsibilities, promote the extension of the river (lake) leadership system to the village level, and explore the establishment of a long-term management and protection mechanism for the treatment of black and odorous water bodies in rural areas.

Strengthen the comprehensive utilization of straw and the supervision of the prohibition of burning. Promote the use of straw as fertilizer, feed, fuel, base material and raw material according to local conditions. Establish a straw resource account system and regular dispatch mechanism to improve the

system, promote deep tillage and green storage, ammoniation and practical technologies, improve the utilization of straw as "five materials", strengthen open-air burning supervision, and strictly implement the burning supervision target responsibility assessment and reward and punishment system.

The utilization rate reached over 94%.

3. Promoting the creation of waste-free cities

Actively promote the construction of "zero waste cities". Learn from the advanced experience of the "zero waste city" pilot projects, connect the "zero waste city" construction indicator system, promote the formation of green development and lifestyle, and coordinate the construction of a green system.

We will continue to promote the reduction of solid waste at the source and resource utilization through the use of clean production processes and the creation of green factories, minimize the amount of landfill, and minimize the environmental impact of solid waste.

Promote the classified collection and treatment of domestic waste. Strengthen the classified collection, source reduction and resource utilization of domestic waste, establish a classification system for the entire process of daily domestic waste delivery, collection, transportation and treatment, and comprehensively

Strengthen the construction of urban and rural domestic waste recycling systems. Fully promote closed collection and transportation, realize dry and wet classification collection and transportation, and implement full-process supervision of domestic waste. Build 9 new county-level domestic waste sorting centers, including 10 first-level

There are 4 second-level sanitation complexes and 61 third-level sanitation complexes. By 2025, the coverage rate of urban domestic waste classification will reach more than 98%.

Strengthen the supervision and management of restaurant kitchen waste. Improve the collection and transportation measures of restaurant kitchen waste, strengthen the supervision and management of restaurant kitchen waste in the city, speed up the construction of comprehensive kitchen waste disposal sites, and ensure the classification of restaurant kitchen waste.

Establish a two-level collection and transportation system for restaurant kitchen waste in Zhengzhou City, namely, the city and district/county (city). Actively compile plans for restaurant kitchen waste treatment facilities, speed up the construction and layout of a restaurant kitchen waste treatment network in the city, and improve the restaurant kitchen waste treatment network.

By 2025, the harmless treatment and disposal rate of restaurant kitchen waste in the city will reach 98%.

Standardize the transportation, disposal and utilization of construction waste. Vigorously promote the on-site and nearby utilization of construction waste. Construction waste must be collected separately and classified in a special way and sent to designated construction waste disposal sites for disposal.

In the construction of parks, ecological heritage parks, ecological corridors (greenways), and water system management projects in cities and towns, it is encouraged to dispose of and utilize construction waste on-site or nearby as needed.

Or reused in foundation reinforcement, road engineering cushion and non-load-bearing concrete hollow blocks, concrete hollow partition boards, autoclaved fly ash bricks, etc. By the end of 2025, the comprehensive utilization rate of construction waste in the city will reach

Strengthen the control of plastic waste sources. Prohibit the production and sale of ultra-thin plastic bags with a thickness of less than 0.025 mm and polyethylene agricultural mulch with a thickness of less than 0.01 mm. Prohibit the use of medical waste as raw materials to manufacture plastics.

The sale of daily chemical products containing plastic microbeads is prohibited. Shopping malls, supermarkets, pharmacies, bookstores and other places, as well as catering take-out services and various exhibitions are prohibited from providing and using non-degradable plastic bags. Express delivery outlets are prohibited from

Plastic packaging bags, plastic tapes, disposable plastic woven bags, etc. Support the construction of a scientific and technological innovation platform for degradable alternative materials, and encourage the development of related industries and production of national-level bio-based degradable materials and alternative products.

Support and encourage the establishment of waste plastic resource utilization enterprises, promote their standardization, centralization and industrialization, increase the construction of plastic waste recycling and disposal capabilities, and reduce the stock of white pollution.

Carry out the construction of the venous industrial park. With the venous industrial park as the center, the layout of waste incineration power generation, restaurant waste treatment, construction waste treatment, sludge treatment, large waste treatment, comprehensive sorting and utilization

Terminal treatment facilities such as wastewater treatment plants, leachate treatment, ash treatment, and slag treatment will be built. Centralized solvent recovery and activated carbon regeneration will be promoted, and pilot projects for the construction of centralized activated carbon regeneration and disposal centers will be encouraged. By the end of 2025,

Construction of venous industrial parks in Xingyang and other places.

4. Pay attention to the prevention and control of noise and vibration pollution

Improve the noise monitoring and supervision system. Adjust the acoustic environment functional zones of Zhengzhou City and its districts and counties (cities) and improve the setting of corresponding monitoring points, promote the construction of an automatic environmental noise monitoring system, and clarify the various types of noise.

Carry out the construction of an assessment and evaluation mechanism for the prevention and control of environmental noise pollution in districts, counties and cities. By 2025, realize automatic monitoring of the acoustic environment quality in functional areas, and achieve a nighttime compliance rate of 85% in urban acoustic environment functional areas.

Strengthen noise management and vibration control. Strengthen traffic noise pollution prevention and vibration control, strictly control social life noise, increase construction noise management and law enforcement, and encourage citizens to actively participate.

Work.

Column 7 Key projects for environmental protection infrastructure construction
1. Central heating reconstruction. Reconstruction project of old primary heating network and primary support network in urban areas, old communities in Zhengzhou Heating facility renovation project, Zhengzhou "West-to-East Heat Transmission" trunk pipeline network project, Xinmi Yuzhong Power Plant "Bringing Heat into Zhengzhou" (Phase I) Long-distance heat transmission network expansion project. 2. Domestic waste treatment. Zhengzhou (South) Environmental Protection and Energy Engineering, Zhengzhou (West) Environmental Protection and Energy Engineering. 3. Sludge and hazardous waste disposal. Zhengzhou Kangning Special General Technology Co., Ltd. Ash Harmless Resource Disposal Project Project, Xingyang 30,000 tons of waste oil filter, oil pot, oil barrel comprehensive disposal project, Municipal Medical Waste Disposal Center Medical Waste Wastewater treatment equipment expansion project, centralized wastewater treatment project for laboratories of Zhengzhou University and other universities, sewage treatment project in Zhengzhou New District The plant is equipped with a 1,000-ton/day sludge resource comprehensive utilization project, and Xingyang City has an annual processing capacity of about 10,000 tons of activated carbon. Regenerative central vein industrial park construction project.

10. Strengthen risk prevention and control to ensure environmental safety

1. Strengthening environmental risk prevention and control and emergency response

Strengthen ecological and environmental emergency prevention, control and management. Strengthen the investigation and assessment of environmental risks in hazardous and serious enterprises, centralized drinking water sources and regions, and implement classified and graded risk management. Strengthen emergency environmental incidents

By the end of 2022, the emergency response plan for environmental emergencies of county-level and above governments will be revised; by the end of 2023, the risk assessment of environmental emergencies in Zhengzhou will be completed; and the middle route main canal of the South-to-North Water Diversion Project (Zhengzhou) will be completed.

Emergency response plan, "one river, one map, one policy" environmental emergency response plan for Jialu River, Shuangli River and Ying River was compiled. Electronic filing of corporate environmental emergency response plans was carried out, and all enterprises involved in hazardous and serious accidents were covered.

Environmental management and special rectification, in-depth investigation of environmental risks and hidden dangers, and establishment of a hazardous waste risk prevention and control mechanism with inter-departmental linkage and regional collaboration. Promote the construction of Zhengzhou Environmental Emergency Material Reserve and improve the emergency material reserve.

Carry out comprehensive treatment of heavy metal pollution. Increase the efforts of non-ferrous metal industry enterprises to upgrade production technology, jointly promote pollution reduction and carbon reduction, and reduce heavy metal pollutant emissions and carbon emissions. Promote electroplating enterprises

Enterprise entry rate. Focus on heavy metal pollutants such as lead, mercury, and cadmium, and study and promote the environmental management of heavy metals throughout their life cycle. Further promote key rivers, lakes, reservoirs, drinking water sources, farmland and other environmentally sensitive areas.

Comprehensive control of industrial pollution.

Strengthen ecological environment and health management. Continue to carry out ecological environment and health literacy improvement activities. Explore and carry out environmental and health surveys in key areas, river basins, and industries, and gradually establish a comprehensive ecological environment and health survey covering pollution sources, environmental

Establish an integrated environmental and health monitoring network and risk assessment system for environmental exposure and health effects. Promote the identification and investigation of ecological and environmental health risks and establish a basic database of ecological and environmental health risk source enterprises.

Include it in the scope of ecological and environmental management, and explore the establishment of a post-evaluation mechanism for sudden environmental incidents and a public health impact assessment system.

(II) Strengthening environmental management of hazardous waste

Improve the capacity for hazardous waste collection, utilization and disposal. Comprehensively carry out pilot work on the centralized collection and storage of hazardous waste from small-scale waste-producing units, promote the construction of regional hazardous waste collection points and storage facilities, and improve the quality of hazardous waste.

Improve the hazardous waste collection and transportation capacity of industrial parks and other places, and improve the hazardous waste collection and transportation system. Improve the county-level medical waste collection, transportation and disposal system and cover rural areas, and encourage the development of mobile medical waste disposal

Provide convenient disposal services at all levels. Strengthen the classification management of medical waste, do a good job of source classification, and promote standardized disposal. Carry out an assessment of the operation status of centralized medical waste disposal facilities, and improve the management of medical waste in major epidemics.

The disposal of medical waste during the epidemic period will be hindered. The supervision of the entire process of hazardous waste will be strengthened. Standardized environmental management of hazardous waste will be promoted, and a system of inter-departmental linkage, regional cooperation, and consultation and supervision of major cases will be established to form a

A supervision system covering the entire process of production, collection, storage, transfer, transportation, utilization and disposal. Carry out the construction and application of the "Internet +" supervision system, and realize the management plan,

The "integration of multiple flows" of declaration, registration, transfer and transportation services. Strengthen environmental law enforcement of hazardous wastes, establish a joint law enforcement mechanism, and severely investigate and punish illegal storage, random dumping and illegal landfill of hazardous wastes in accordance with the law.

Strengthen the control of new pollutant emissions. Strengthen the environmental risk control of new pollutants in industries such as coatings, textile printing and dyeing, rubber, and medicine. Strengthen the environmental management registration and supervision of new chemical substances, and strengthen in-process and post-process supervision.

Environmental risks of new chemical substances that are persistent, bioaccumulative, and harmful to the environment and health. Fully implement the elimination and restriction measures for toxic and hazardous chemical substances in the Guiding Catalogue for Industrial Restructuring, and strengthen

Promote the application of alternative technologies. Implement mandatory clean production audits in accordance with the law for enterprises that use toxic and hazardous chemicals or emit toxic and hazardous chemicals during production.

Accelerate the elimination, restriction and reduction of chemicals managed and controlled by international environmental conventions. Eliminate hexabromocyclododecane, decabromodiphenyl ether, perfluorooctylsulfonic acid and its salts and perfluorooctylsulfonyl fluoride, and basically eliminate short-term

Support the research, development and application of mercury-free catalysts and processes, and encourage the research and application of alternatives and technologies for restricted or banned persistent organic pollutants.

Illegal production and use, illegal production of mercury-added products and other illegal activities.

3. Improving the level of radiation pollution prevention and control

Strengthen radiation safety supervision. Continue to optimize the supervision mechanism and improve the safety supervision system of radioisotopes and radiation devices. Strengthen the supervision of radiation construction projects during and after the project, and promote the construction of radiation safety.

Strengthen the supervisory management of key radiation pollution sources and electromagnetic radiation facilities, and urge nuclear technology users to establish and improve long-term radiation safety mechanisms. Ensure that waste and idle nuclear power equipment is properly used.

All collected and stored.

Promote radioactive pollution control. Strengthen the safety supervision of radiation environment in the development and utilization of associated radioactive mines, and urge enterprises included in the provincial list of associated radioactive mine development and utilization to conduct radiation environment and effluent safety inspections on their own.

Disclose to the public in a timely manner.

11. Enhance regulatory capabilities and build a modern environmental governance system

1. Improve the ecological environment management mechanism

Improve the leadership responsibility system for environmental governance. Strictly implement the leadership responsibilities of the Party and government, and insist on the same responsibility of the Party and government and dual responsibility for one post. Improve the overall responsibility of the municipal party committee and municipal government, and the management committees of each development zone, district, county (city) and municipal government.

Implement the ecological civilization construction goal evaluation and assessment system, and use the assessment results as an important basis for evaluating the performance of leading cadres, annual assessment and selection and appointment. Fully implement the natural resource resource management system for leading cadres.

We will strictly implement the system of lifelong accountability for ecological and environmental damage, implement rectification work on issues reported by the central and provincial ecological and environmental protection inspections, and improve the inspection and rectification promotion mechanism.

Strengthen the main responsibility of enterprises for environmental governance. Implement the main responsibilities of enterprises for pollution control, damage compensation and ecological restoration, promote the greening of enterprise production and services, and enhance the greening management of industrial products throughout their life cycle.

Extended responsibility system. Reduce resource consumption and pollutant emissions from the source, eliminate backward production processes, and practice green production methods. Implement a green environmental dispatch system and continue to carry out energy efficiency and water efficiency leaders.

action.

Strengthen environmental protection and social supervision by the whole people. Promote the construction of the "12369" environmental protection hotline platform, improve the system of accepting, investigating, responding to and rewarding public reports, strengthen public opinion supervision, and encourage news media to

Improve the public participation mechanism in environmental decision-making and safeguard the public's right to know, right to supervise and right to participate. Give full play to the role of various social groups and vigorously improve

(II) Give full play to the incentive effect of market mechanism

Improve the environmental governance market system. Cultivate and standardize the environmental governance market, improve the environmental protection reward and punishment mechanism, actively promote third-party governance of environmental pollution, further optimize the business environment, and stimulate market players.

Governance market system. Carry out pilot projects on comprehensive environmental governance trusteeship services in small towns, support overall environmental governance solutions, environmental stewardship, and regional integrated service models. Explore third-party governance units' pollution control

The policy encourages the implementation of "environmental restoration + development and construction" on industrially polluted sites.

Establish an environmental rights trading market. Explore the initial allocation of carbon emission rights, pollution emission rights, energy use rights, and water rights, and improve the market-oriented and diversified ecological protection compensation mechanism. Implement the pollution emission rights trading system.

Trading work. Further improve the regional environmental quality ecological compensation mechanism.

Promote ecological and environmental price reform. Improve the green price mechanism, strictly implement the policy orientation of "whoever pollutes, pays", establish and improve the "polluter pays + third-party governance" mechanism, and implement differentiated

Improve and implement the charging policy for urban sewage and garbage treatment and the charging mechanism for medical waste disposal, and explore the establishment of a charging system for rural domestic sewage and garbage treatment. Implement electricity consumption fees for the environmental protection industry.

Determine the price of recycled water.

3. Improve the legal and regulatory management system

Strengthen the legal protection of the ecological environment. Improve local laws and regulations on the ecological environment, increase research on the legislative needs for resource conservation and utilization, pollution prevention and control, and biodiversity protection, and promote soil pollution prevention in Zhengzhou.

We have carried out legislative revision of local laws and regulations such as the Measures for the Prevention and Control of Environmental Noise Pollution and the Measures for the Administration of Environmental Monitoring in Zhengzhou, and continuously improved the system of laws and regulations on the ecological environment.

Strictly implement the pollutant discharge permit system. Achieve full coverage of pollutant discharge permits for fixed pollution sources in the city and establish a dynamic management mechanism for pollutant discharge permits. Implement the "one-permit" supervision system to promote the integration of pollutant discharge permits and the ecological environment.

Effective connection of law enforcement, environmental monitoring, total volume control, environmental impact assessment and other systems. Establish a normalized supervision and law enforcement mechanism based on pollutant discharge permits. By the end of 2024, the pollutant discharge permit system will be implemented.

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The linkage between detection, law enforcement and supervision is smooth.

Improve the environmental credit system. Improve the Zhengzhou enterprise environmental credit information platform and implement the legal disclosure system of environmental information. Establish a blacklist system for polluting enterprises and improve the collection and evaluation of environmental credit information.

Establish a credit punishment mechanism and build a hierarchical and differentiated credit supervision model based on environmental credit rating. Explore the establishment of environmental service institutions such as environmental impact assessment report preparation, environmental testing services, and third-party environmental governance.

Improve the punishment and exit mechanism for the environmental services industry.

Strengthen judicial linkage for environmental protection. Implement information sharing, case notification, and case transfer systems among comprehensive administrative and law enforcement agencies, public security agencies, procuratorates, and judicial agencies for ecological and environmental protection. Promote environmental

Link administrative penalties, criminal justice and ecological and environmental damage compensation systems. Increase the intensity of investigation, prosecution and trial of ecological and environmental crimes. Implement the ecological and environmental damage compensation system

Screening, tracking and handling of major cases, and evaluation of restoration effectiveness.

(IV) Improving ecological and environmental supervision capabilities

Strengthen the construction of comprehensive law enforcement capabilities. Implement the "bureau-team integration" system, follow the principle of local management and downward focus, focus on strengthening law enforcement at the city and county levels, and strive to improve grassroots law enforcement and supervision capabilities.

Incorporate environmental protection related work into daily law enforcement content. Promote the standardization of comprehensive law enforcement agencies, equip them with equipment such as investigation and evidence collection, mobile law enforcement, and data analysis, ensure the use of vehicles for front-line ecological and environmental law enforcement, and accelerate

We will deepen the "double random, one public" environmental supervision model, implement the positive list of supervision and law enforcement, carry out special law enforcement, cross-enforcement and cross-departmental joint enforcement, and strengthen the supervision and law enforcement of the "double random, one public" environmental supervision model.

Make full use of the "Internet + supervision" system, satellite remote sensing, drones, drones, navigation vehicles, big data analysis and other scientific and technological means to carry out "non-site" monitoring and "non-contact" law enforcement.

Tube.

Strengthen environmental monitoring capacity building. Continue to improve the monitoring network covering air, water, soil, noise, radiation and other environmental factors, enhance the basic ecological monitoring capacity and ecological remote sensing monitoring capacity, and comprehensively

Carry out carbon monitoring and assessment pilot projects, establish VOCs automatic monitoring stations in VOCs-related industrial clusters and industrial parks, build traffic pollution monitoring stations, and gradually build

The three-dimensional monitoring network will be jointly controlled. The construction of new municipal control assessment section sites in the Yellow River Basin will be promoted to enhance the automatic monitoring and control capabilities of the Yellow River Basin water quality. In accordance with the relevant requirements of the state and the province, the "Fourteenth Five-Year Plan" national land

The groundwater environment monitoring network will be improved with the groundwater pollution tracking monitoring wells as the main monitoring wells, and the groundwater environment quality investigation and assessment monitoring wells in the central urban area of Zhengzhou as supplements. Improve the pollution source monitoring system

Strengthen the rural ecological environment monitoring network, conduct regular monitoring of farmland irrigation water in large and medium-sized irrigation areas, deepen monitoring quality management and comprehensive data utilization, and improve environmental monitoring.

Management system. By 2025, monitoring agencies in all districts, counties and cities should meet relevant national standardization construction requirements.

Strengthen ecological protection law enforcement and supervision. Conduct basic surveys of nature reserves, accelerate the construction of ecological protection red line supervision platforms, and strengthen the area, function, nature and management implementation of ecological protection red lines.

Carry out ecological protection red line monitoring and early warning. Strengthen supervision and law enforcement in key areas such as ecological protection red lines and nature reserves, regularly carry out regular law enforcement supervision and inspection of ecological protection red lines, and continue to carry out the "Green Shield" natural

Supervision, timely detection of various ecological damage behaviors and follow-up supervision.

Improve the ecological and environmental emergency response capabilities. Focus on tailings ponds, smelting enterprises, etc., improve the prevention and mitigation of sudden ecological and environmental incident risks and emergency preparedness responsibility system, and strictly implement the main responsibility of enterprises.

Strengthen the configuration of emergency monitoring equipment, improve the multi-level environmental emergency expert management system, carry out rotation training for environmental emergency personnel in a classified and graded manner, and enhance

Establish a nuclear-related social risk warning and public opinion control mechanism, revise and improve Zhengzhou's radiation accident emergency plan, and enhance radiation emergency response monitoring capabilities.

(V) Strengthen the construction of scientific and technological support system

Strengthen the ability of ecological and environmental scientific and technological innovation. Strengthen the basic capabilities such as technology research and development in the field of ecological environment, rely on the resource advantages of universities and research institutes, establish a number of key laboratories and engineering technology centers, and promote

Independent product innovation and achievement transformation. Focus on promoting the coordinated reduction of nitrogen oxides and volatile organic compounds, coordinated control of PM2.5 and O3, coordinated efficiency enhancement of pollution reduction and carbon reduction, ecological protection and restoration, and clean energy utilization.

Promote the research and development and application of advanced technologies such as environmental monitoring, pollution prevention and control technologies, and equipment. Promote the application of scientific and technological innovation in the field of environmental monitoring. Promote the transformation of scientific and technological achievements and improve the scientific and technological achievements of environmental protection.

Improve statistical and evaluation capabilities. Strengthen the basic capacity building of ecological and environmental statistics, focus on business training, continuously improve the quality of ecological and environmental statistical data, and promote the standardization and regulation of ecological and environmental statistical work.

Continuously improve the level of ecological environment statistical management and carry out basic research on ecological environment statistical parameters. Explore and carry out various types of ecological environment governance performance evaluation, carry out rural domestic sewage treatment effectiveness evaluation and operation and maintenance

Study.

Promote the informatization of ecological and environmental governance. Strengthen the integration and improvement of data resources, establish an ecological and environmental data sharing mechanism, and improve the level of smart environmental protection. Establish a unified database of fixed pollution sources in Zhengzhou to achieve

Build an "integrated picture" of ecological and environmental informationization. Accelerate the intelligent upgrade of the Yellow River ecological and environmental protection facilities and build the Yellow River Ecological Big Data Center. Strengthen the construction of ecological and environmental business application systems and promote the integration of various business application systems.

Provide support for environmental intelligent business applications. Make full use of big data, cloud computing, blockchain, artificial intelligence and other means to promote digital pollution control and scientific pollution control, and achieve rapid and accurate perception and intelligent environmental pollution control.

Safety management, improve the network security responsibility system and information security technology protection system. Strengthen the sharing of data such as agricultural source census, ecological environment statistics, comprehensive utilization information of livestock and poultry manure, and pollution discharge permits. Cooperate with the provincial

Construction of pollution supervision platform.

Column 8 Key projects for improving ecological and environmental governance capabilities
1. Ecological and environmental law enforcement and supervision capacity building project. According to the standard configuration of law enforcement vehicles, law enforcement evidence collection equipment and Office equipment, law enforcement clothing and protective equipment, construction of drone monitoring platform, purchase of road inspection law enforcement vehicles OBD detection system equipment, carry out business competency training, and establish a teacher and talent pool for environmental law enforcement. 2. Ecological environment intelligent perception monitoring capability construction project. Construction of roads, airports, railway freight yards and other transportation pollution There are 1 pollution monitoring station each, and 1 monitoring station for the equipment industry cluster in Shangjie District and the automobile and parts manufacturing enterprise cluster in Zhengzhou Economic Development Zone. Dazhongyuan Automobile City and the furniture enterprise clusters in Guodian Town and Changhu Town of Xinzheng City, as well as industries with high nitrogen oxide emissions. Conduct collaborative monitoring of industrial clusters and industrial parks, and select 2-3 industrial parks to carry out aerial monitoring of VOCs pollution. Build the Zhengzhou Songshan greenhouse gas background monitoring station. Improve the air pollutant source tracking flight monitoring system and implement VOCs Automatic monitoring network construction, purchase of monitoring sampling vehicles, sampling and analysis equipment, configuration of intelligent air and water monitoring equipment

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Build a high-altitude remote sensing inspection platform for pollution sources and establish dynamic information management of rivers and lakes

system to build an intelligent wastewater online monitoring platform.

3. Ecological environment information construction project. Build a smart environmental protection big data platform.

12. Implement key projects and establish a mechanism to ensure planning implementation

1. Clarify the division of responsibilities

All district, county and city governments should incorporate the goals and tasks set out in this plan into their own government goal assessments, and formulate and announce annual goals and key tasks for ecological and environmental protection.

Regular communication will be conducted to resolve major issues encountered during the implementation of the plan and ensure the completion of the planning objectives and tasks.

(II) Implementing key projects

Implement major projects such as green development, addressing climate change, blue skies, clear waters, clean land, urban and rural coordination, ecological protection and restoration, environmental risk prevention and control, and improving ecological and environmental governance capabilities, and establish plans

Actively apply for the special fund for water and land and the special fund for ecological protection of the Yellow River Basin, and accelerate the implementation of key projects. Support the ecological environment-oriented development of the Yellow River Ecological Comprehensive Demonstration Zone (E

Promote the formation of a scalable and replicable regional ecological environment governance and industrial integration development model.

(III) Increase investment

Implement the division of fiscal powers and expenditure responsibilities at all levels in the field of ecology and environment, improve the fiscal system in the field of ecology and environment, and enhance the ability to ensure basic public services for ecology and environmental protection at the grassroots level. Broaden financing channels,

Guide social capital investment through planning, finance, and pricing policies. Promote cooperation between the government and social capital to attract social capital to participate in quasi-public welfare and public welfare environmental protection projects. Encourage social capital

Establish an environmental protection fund. Encourage venture capital companies, equity investment companies and social donations to increase investment in ecological and environmental protection.

(IV) Strengthen publicity and guidance

Strengthen publicity and education on ecological civilization and environmental protection, incorporate ecological civilization into the national education system, vocational education system and party and government leadership training system, and guide the whole society to raise awareness of ecological civilization and environmental protection.

Strengthen the creation and dissemination of ecological and environmental cultural publicity products. Actively carry out publicity and exchange of ecological civilization construction and ecological and environmental protection planning policies, laws and regulations, progress and achievements, and practical experience.

We will publicize typical demonstrations such as the protection of canal sites and demonstration cities and counties for ecological and civilized construction, and promote advanced experiences and practices. We will also discover the outstanding deeds of a group of advanced individuals and groups, and do a good job in typical reports.

(V) Promoting team building

Strengthen the construction of ecological and environmental protection teams, especially the introduction and training of urgently needed talents in environmental management such as climate change, soil, solid waste, and chemicals; strengthen the construction of ecological and environmental protection teams in districts, counties (cities), towns (streets), and townships (cities).

Strengthen the ecological protection force; improve business capabilities through business training, competitions, experience exchange and other methods. Strengthen the ecological protection of natural resources, agriculture, water conservancy, forestry and other departments responsible for ecological environment protection.

Promote the construction of ecological civilization and ecological environment scientific research capabilities, innovation platform construction, and the training of leading talents and technical experts. Increase the recognition of advanced collectives and individuals in ecological and environmental protection.

6. Conduct implementation evaluation

The Municipal Ecological Environment Bureau, together with relevant departments, will conduct mid-term and final evaluations on the implementation of this plan in 2023 and 2025 respectively.

Attachment: 1. Air quality assessment targets for counties and cities during the 14th Five-Year Plan period

2. Water quality targets for national, provincial and municipal surface water sections during the 14th Five-Year Plan

3. Assessment targets for centralized drinking water sources during the 14th Five-Year Plan

attachment1

Air quality assessment targets for counties and cities during the 14th Five-Year Plan

Serial number	area	Indicator name	2020	2
1	Zhongmu County	Annual average concentration of fine particulate matter (ÿg/m3)	48	
		Number of days with good air quality (days)	256	
2	Xinmi City	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	253	
3	Xingyang City	Annual average concentration of fine particulate matter (ÿg/m3)	53	
		Number of days with good air quality (days)	234	
4	Xinzheng City	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	244	
5	Dengfeng City	Annual average concentration of fine particulate matter (ÿg/m3)	48	

		Number of days with good air quality (days)	260	
6	Gongyi City	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	213	
7	Zhengzhou Airport Economic Zone	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	238	
8	Upper District	Annual average concentration of fine particulate matter (ÿg/m3)	52	
		Number of days with good air quality (days)	223	
9	Huiji District	Annual average concentration of fine particulate matter (ÿg/m3)	52	
		Number of days with good air quality (days)	234	
10	Golden Waters	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	227	
11	Zhengzhou Hi-tech Zone	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	222	
12	Zhongyuan District	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	199	
13	District 27	Annual average concentration of fine particulate matter (ÿg/m3)	51	
		Number of days with good air quality (days)	233	
14	Guancheng Hui District	Annual average concentration of fine particulate matter (ÿg/m3)	50	
		Number of days with good air quality (days)	214	
15	Zhengzhou Economic Development Zone	Annual average concentration of fine particulate matter (ÿg/m3)	50	
		Number of days with good air quality (days)	233	
16	Zhengdong New District	Annual average concentration of fine particulate matter (ÿg/m3)	50	
		Number of days with good air quality (days)	210	

Annex 2

Water quality targets for national, provincial and municipal surface water sections during the 14th Five-Year Plan

Serial number	Section Name	Water body	Assessment Area	The 14th Five-Year Plan Goals
1	Garden Entrance	Yellow River	Zhengzhou City	ÿ
2	Gongyi Qilipu	Yiluo River	Gongyi City	ÿ
3	Jiangang Reservoir	Jialu River	District 27 (Municipal Water Resources Bureau)	ÿ
4	Zhongmu Chenqiao	Jialu River	Zhongmu County	ÿ
5	Xinzheng Huangfu Village	Shuangli River	Xinzheng City	ÿ
6	Baisha Reservoir	Ganghe	Dengfeng City	ÿ
7	Old Village Shangcun	Meihe	Zhengzhou Airport Economic Zone	ÿ

8 Liangjia Bridge	Zhangbagou	Zhengzhou Airport Economic Zone	ÿ	
9. Ma'an Cave	Shuangli River	Xinmi City	ÿ	
10 ÿÿ	Sishui River	Xingyang City	ÿ	
11 Boundary River	Shuangli River	Dengfeng City	ÿ	
12 Seventeen River into Zhengchu	Shiqili River	Xinzheng City	ÿ	
13 Chaohe River Enters Zhengzhou	Chaohe	Xinzheng City	ÿ	
14 Where the Suo River meets the Xu River	Sohe	Xingyang City	ÿ	
15 Xuhe Baxian Bridge	Sukawa	Zhongyuan District	ÿ	
16 Xuhe Expressway Bridge	Sukawa	Zhengzhou Hi-tech Zone	ÿ	
17 Shangjie District Entrance to Xingyang	Dry River	Upper District	ÿ	
18 Kuhe River Xingyang enters Huiji District	Dry River	Xingyang City	ÿ	
19 Where the Dry River Flows into the Yellow River	Dry River	Huiji District	ÿ	
20 Where the Sishui River enters Xingyang from Gongyi	Sishui River	Gongyi City	ÿ	
21 Where Sishui River enters Xingyang Street	Sishui River	Xingyang City	ÿ	
22 Where Sishui River enters Xingyang from Shangjie	Sishui River	Upper District	ÿ	
Note: The annual target is subject to the target issued by the superior department each year.				

Annex 3

Assessment targets for centralized drinking water sources during the 14th Five-Year Plan

No.	Drinking water source protection area	Responsible unit	Category	Water Source Level	Water
1	Huanghe Garden Exit	Zhengzhou City	River	city level equals or exceeds	
2	Huanghe Mangshan	Zhengzhou City	River	city level equals or exceeds	
3	Urban well water plants (27 wells in total)	Zhengzhou City	Groundwater	City-level reaches or exceeds	
4	Jiuwutan underground water well group (36 wells in total)	Zhengzhou City	Groundwater	City-level reaches or exceeds	
5	Jiangang Reservoir	Zhengzhou City	Lake-reservoir	city level reaches or exceeds	
6.	Yellow River Village	Xingyang City	River	county level reaches or exceeds	
7	Xinmi City underground water well group (a total of 11 wells)	Xinmi City	Groundwater	County Level	Meet or exceed outside)
8	Xinzheng Water Plant Underground Water Well Group (5 wells in total) well)	Xinzheng City	Groundwater	County-level reaches or exceeds	
9	Gongyi Sanshuichang underground water well group (a total of 10 wells well)	Gongyi City	Groundwater	County-level reaches or exceeds	
10	Wulao Reservoir	Gongyi City	Lake-reservoir	county-level reaches or exceeds	
11	South-to-North Water Diversion Project Middle Route Main Canal	Xinzheng City, Xingyang City, Zhongyuan District, No. 27 District, Zhengzhou Economic Development Zone, Guancheng Hui District, Zhengzhou Hi-tech Industrial Development Zone District, Zhengzhou Airport Economic Zone	River\	Meets or exceeds	

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